

AI Headstamp Sorter Application Guide

For software version 1.3.8

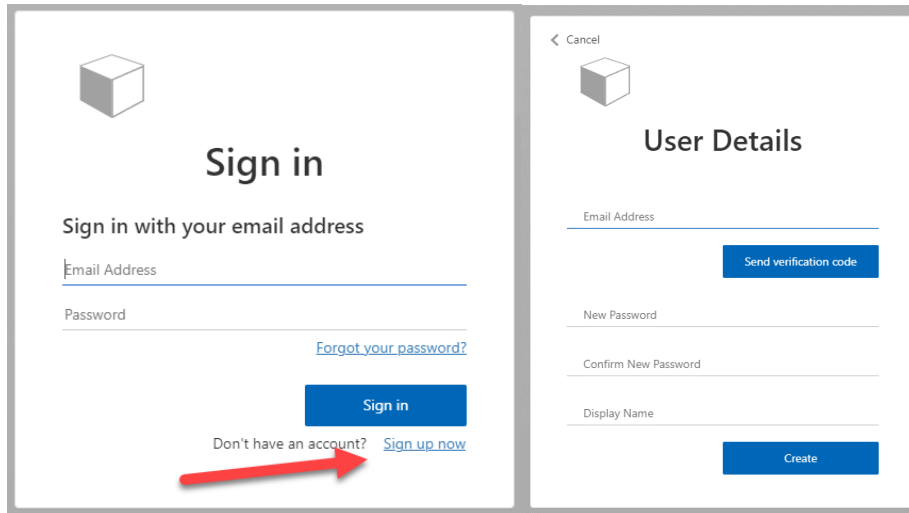
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Download and Install

To download the software, visit <https://www.reloadingsrecipes.com/HeadstampSorter> and click the link to go to the downloads page. You will be required to create an account if you don't have one already. To create an account, click the **Sign up now** button. Minimal information is required to create an account though your email address should be valid to activate the account.



The image displays two side-by-side screenshots of a web application's user interface. The left screenshot is the 'Sign in' page, featuring a cube icon, the title 'Sign in', and the instruction 'Sign in with your email address'. It includes input fields for 'Email Address' and 'Password', a 'Forgot your password?' link, a 'Sign in' button, and a 'Sign up now' link. A red arrow points to the 'Sign up now' link. The right screenshot is the 'User Details' page, also featuring a cube icon and the title 'User Details'. It includes input fields for 'Email Address', 'New Password', 'Confirm New Password', and 'Display Name', along with a 'Send verification code' button and a 'Create' button.

A note on Privacy and Data Protection and Internet Connectivity

Though there is not an official privacy policy, the only data being collected is the account information you input when creating the account which is Email, Password & Display Name. Passwords are stored in one-way hash and cannot be decrypted. The email is validated on initial account creation.

The application requires sign-in to run initially and automatically generates an offline license based on the machine key. The offline license is good for a year after which time, you would be required to sign-in again to generate a new license.

Pre-Installation

Once you have created your account and signed in, you can download the software by clicking the link to the latest version. The installation will require approximately 8GB of disk space to operate. You will need more space as you download or create additional models. Having 20-30GB of space available is recommended.

It is a good idea to also ensure that you have the Microsoft VC++ Redistributable installed as well. Most systems will have this installed already but if you skip this step and run into issues installing the software, you might have to come back and install the redistributable.

Important! - This app requires Microsoft VC++ Redistributable to be installed

You may already have it installed as many apps require it but if you are using a new installation of windows you need to install this first.

- [\(Direct Download\) Microsoft VC++ Redistributable x64](#)
- [Microsoft VC++ Redistributable Page](#)

[AI Case Sorter Installer \(v1.1.42\) 2/3/2024](#)

Requires Arduino Firmware version 7.1.240130.1 or later!

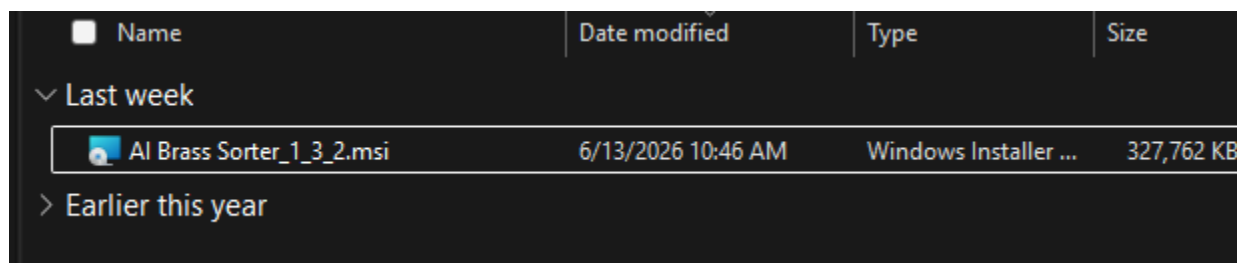
Features

- Added option to enable/disable auto add-feed on headstamp count click in training screen
- Added multi-image classification processing mode to switch between serial and parallel processing

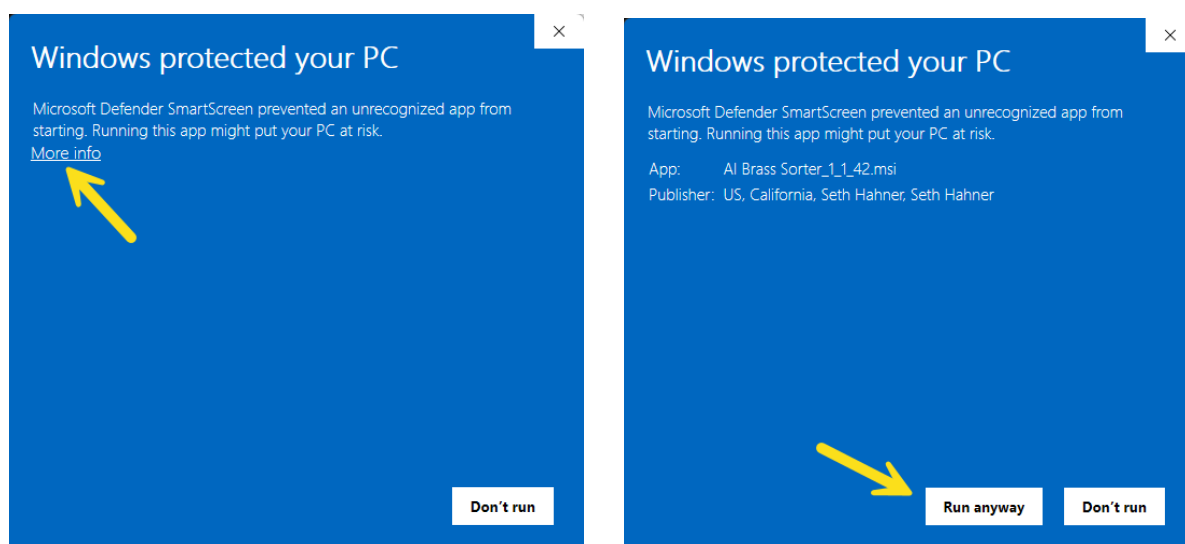
Bug Fixes

- Fixed issue where machine cycles twice on headstamp count click
- Updated run screen Automatically Select Trays to respect confidence floor.

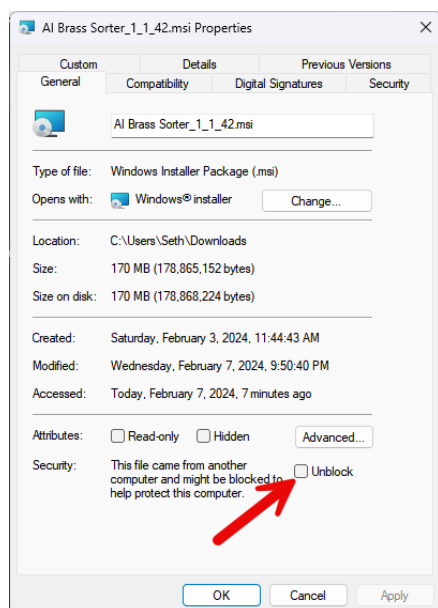
Once your download is complete, you can launch the installer to begin the installation. The installer should look something like the following screenshot but the version number will be different.



After launching the installer, you may receive a notification from Windows Defender SmartScreen as seen in the following screenshot. If so, click on the “More Info” link and then you will be given the option to install anyway.

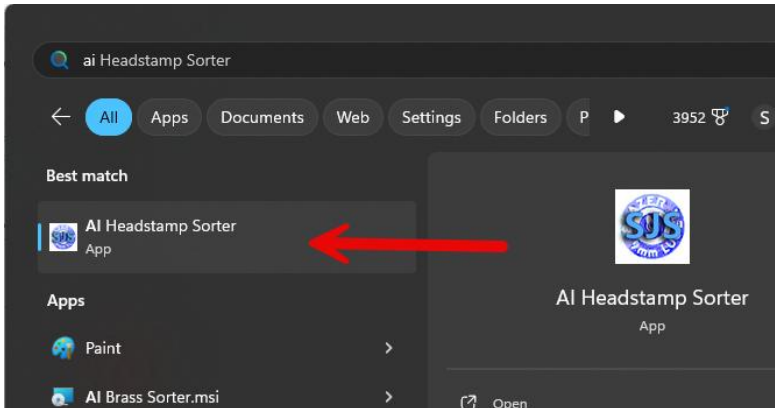


Alternatively, you can close this window and then right click on the file and go to properties and check the “unblock” check box. After unblocking, you should be able to run the file without the above messages.

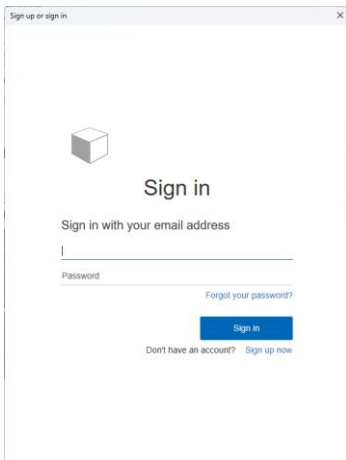


Once you have completed the installation, you can launch the application by going to your start menu and searching for “AI Headstamp Sorter”.

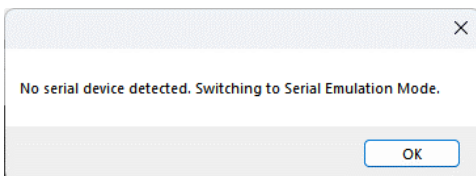
Starting the Software



The first time the software is launched, you need to have an internet connection. You will see a prompt that requires that you sign-in again with the same account you used when you downloaded the software. The application may reach out to the internet initially to download additional assets required for the machine learning elements (usually seen during the first training). It is recommended before air-gapping (removing network connection), that you sign-in, create a basic model with a couple classifications (in emulator mode) and then train the model. You probably want to do this twice (once with deep learning enabled and once without). After that point, the network cable/wifi can be “unplugged” and the system will operate in offline mode for a year unless the hardware internally is changed significantly.



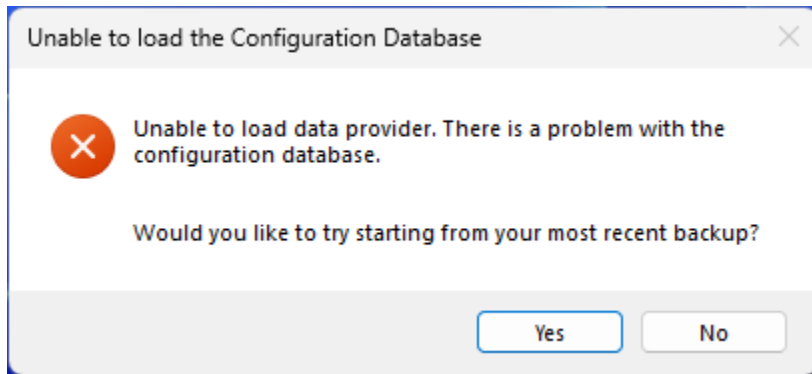
Once you have completed the sign in process, the software will launch. If the sorter hardware is not yet connected to the computer (via USB cable), the software will launch in “Emulation Mode” and you will see a prompt indicating so.



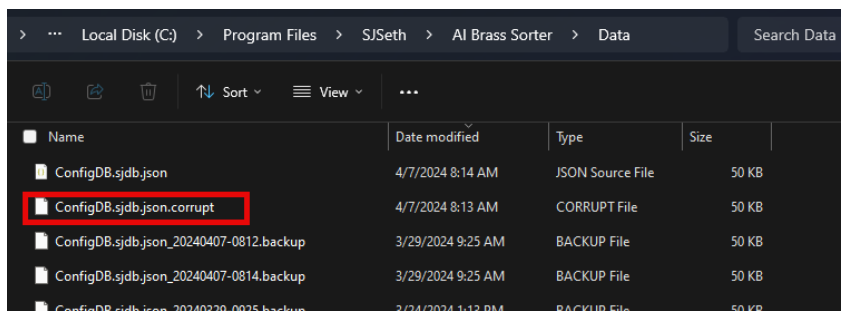
The software also requires that you have a web camera attached to the system even to run in emulation mode. If you do not have a camera, you will see an error message indicating it is required and the software will close.

Corrupted Data File Exception

In the unlikely event that your configuration data file has become corrupted, as of version 1.1.47, the software will prompt you of the event and ask if you would like to start from the latest backup.



If you select no, the software will exit. If you select yes, the data file will be renamed with the extension “.corrupt” and be replaced by the latest backup.



Backups are taken automatically whenever a configuration change is saved. The last 10 backups are retained.

Initializing Environment (Python Setup)

When the software is started for the first time, you will see a popup message requesting you to indicate the mode used for PyTorch. Unless you have an modern Nvidia graphics card, select the CPU only mode which is the default setup. If you have a compatible graphics card, you will benefit by significantly reducing the time takes to train a model. The following is list of known working graphics cards but yours may work even though not in the list:

- NVIDIA RTX 3050 - 3090
- NVIDIA RTX A2000 – A6000
- NVIDIA RTX 4060 – 4090
- NVIDIA RTX 2000 Ada – 6000 ada
- NVIDIA RTX 5060 – 5090
- NVIDIA RTX PRO 2000 - 4000

Choose PyTorch Setup Mode

PyTorch setup is required

Select how you want to install PyTorch for this machine:

Use CPU Only (Recommended for most users)

Works on all systems. Recommended for most users.

Use NVIDIA GPU

Requires NVIDIA GPU with Compute Capability 8.0+ and at least 4 GB VRAM

Once you have selected your installation type, you will see a status dialog while python and pytorch is configured.

***Note: An internet connection is required for this installation.**

Because pytorch is a very large installation, this stage may take anywhere from 5 minutes to hours depending on your internet connection speed and computer performance. The installer should automatically detect and install the correct version of PyTorch and TorchVision for your computer. The entire installation of pytorch will take approximately 6GB of disk space so ensure you have adequate space.

Getting Started

On successful launch, the software will open into the Configuration panel.

The areas boxed in red from the screenshot below are “global” and will be seen on nearly all pages of the software. On the left, there are the navigation buttons, at the bottom is the current application and firmware version and in the top right corner is the currently selected Cartridge and Model. There is a default empty model created as seen in the screenshot.

Headstamp Sorter

Licensed as: Seth H Gmail(sethshahner@gmail.com)

Configuration

Image Processing

Train

Run

Models

Community

AIConfig

Serial Communication

Select Port COM4 Rescan Ports

Test Feed Test Sorter Slot 0

Home Feed Home Sorter

Feed Motor Speed 94

Feed Cycle Steps 60

Feed Homing Offset 5

Motor Standby (sec) 0

Debounce Timeout 500

Debounce Pause Time 500

Feed Motor Current 1000

Sort Motor Current 1000

Sort Motor Speed 94

Slot Step Inc 20

Sort Homing Offset 0

Slot Drop Delay (ms) 300

Notification Delay (ms) 120

Slot Count 8

AirDrop Configuration

☐ AirDrop Enabled

Pre Drop Delay (ms) 0

Signal Duration (ms) 0

Post Drop Delay (ms) 100

Cartridge: 9mm

Model: AIServerLocal



Save Settings

☐ Initialize Settings on Startup

Restore Defaults

Version: 1.3.2.0

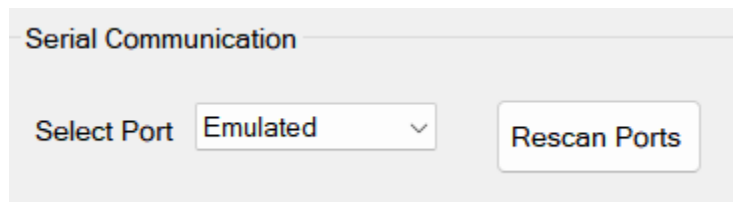
Firmware: 7.2.260128.7.1

There are 7 main areas in the software and corresponding navigation buttons.

- **Configuration** – This screen is used to control the configuration for the application and the runtime Arduino settings.
- **Image Processing** – This is where you can adjust your camera and capture settings as well as your captures modes and brass detection settings.
- **Train** – The training screen is where you capture training images when creating a new machine learning model. A model must be created, have images added to it and be trained before any sorting can be done
- **Run** – The run screen is where you do the actual sorting. Once you have created and trained a model, you will select how you want things to be sorted and start the sorting process.
- **Models** – The models screen is where you can create and manage your cartridges and corresponding machine learning models. By default, the 9mm cartridge has been created along with a default empty model. In emulation mode, the emulated pictures are captured from 9mm brass and you can experiment with adding and classifying images into the default model.
- **Community** – The community tab is only available if you are connected to the internet. This tab will show you those community created and managed models which are available for download.
- **AI Config** – This tab will only be shown if the current model you have selected is an OpenAI API Model. When selected, the train tab will be grayed out as it is not applicable for these model types.

Configuration Screen

Serial Communication

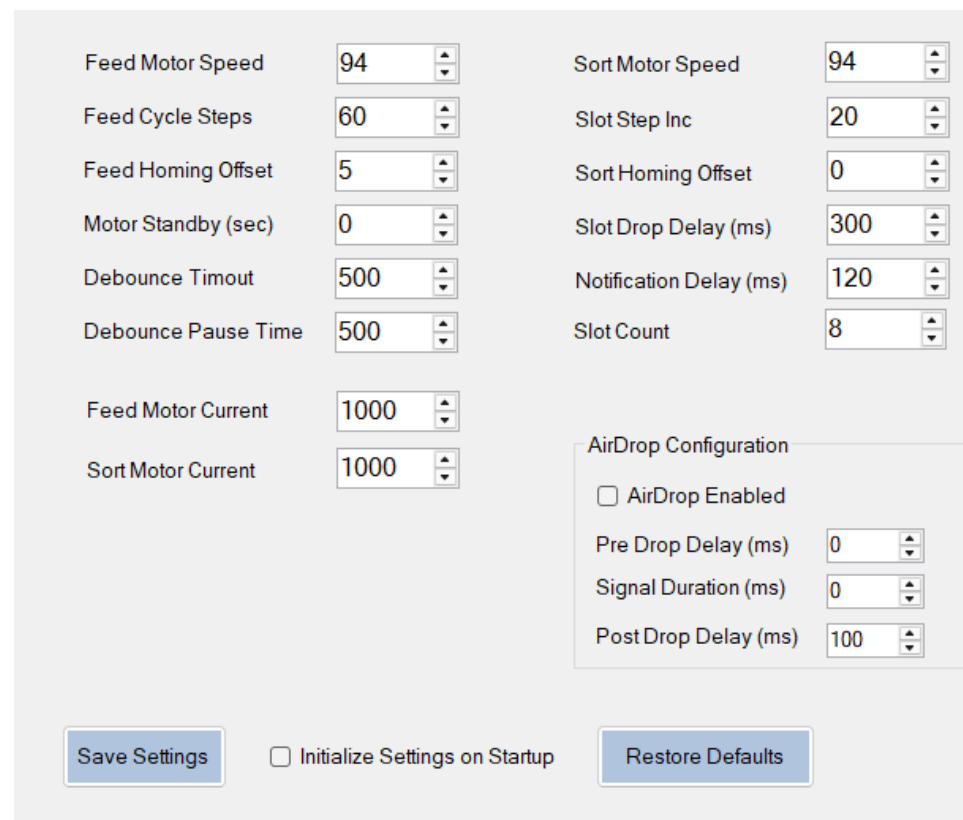
The image shows a configuration window titled "Serial Communication". It contains a "Select Port" label followed by a dropdown menu currently showing "Emulated". To the right of the dropdown is a button labeled "Rescan Ports".

Serial Communication

Select Port Emulated Rescan Ports

This section allows you to select from a list of detected devices. If your sorter is connected, you should see a COM port present in this list. If you do not, try disconnecting the USB cable and reconnecting it and then click the “Rescan Ports” button. If it detects the device, it will automatically be selected and visible in the list. If not, the application will remain in Emulation mode.

Hardware Configuration Settings

The image shows a configuration window titled "Hardware Configuration Settings". It contains two columns of settings, each with a label and a numeric input field with up/down arrows. The settings are: Feed Motor Speed (94), Feed Cycle Steps (60), Feed Homing Offset (5), Motor Standby (sec) (0), Debounce Timeout (500), Debounce Pause Time (500), Feed Motor Current (1000), Sort Motor Current (1000), Sort Motor Speed (94), Slot Step Inc (20), Sort Homing Offset (0), Slot Drop Delay (ms) (300), Notification Delay (ms) (120), and Slot Count (8). There is also an "AirDrop Configuration" section with a checkbox for "AirDrop Enabled" (unchecked) and three numeric input fields: "Pre Drop Delay (ms)" (0), "Signal Duration (ms)" (0), and "Post Drop Delay (ms)" (100). At the bottom, there are three buttons: "Save Settings", "Initialize Settings on Startup" (with an unchecked checkbox), and "Restore Defaults".

Feed Motor Speed 94 Sort Motor Speed 94

Feed Cycle Steps 60 Slot Step Inc 20

Feed Homing Offset 5 Sort Homing Offset 0

Motor Standby (sec) 0 Slot Drop Delay (ms) 300

Debounce Timeout 500 Notification Delay (ms) 120

Debounce Pause Time 500 Slot Count 8

Feed Motor Current 1000

Sort Motor Current 1000

AirDrop Configuration

☐ AirDrop Enabled

Pre Drop Delay (ms) 0

Signal Duration (ms) 0

Post Drop Delay (ms) 100

Save Settings ☐ Initialize Settings on Startup Restore Defaults

These settings can be used to override the default settings on your Arduino firmware. They will only be applied temporarily and will be reset when the Arduino resets or is powered off. To ensure the settings are always used, you need to check the checkbox titled “Initialize these Settings on Startup”.

Here is a summary of each of the settings:

- **Feed Motor Speed** – range of 1 to 100 which controls how fast the feed motor turns. A speed of around 90 is usually preferred here.
- **Feed Cycle Steps** – This setting is mostly an artifact from when the sorter did not have a homing mechanism. If a homing switch is connected, this setting should be set to 70. If a homing switch is not connected, this setting should be set to 80 as that is the number of steps between each position in the feed

wheel. The setting represents the number of “blind” steps the motor should run before looking for the homing switch signal.

- **Feed Homing Offset** – This is the number of additional steps the motor should turn after the homing switch has been triggered. It can help with centering your brass over the camera but you must be careful not to use too large a value here as it can cause the system to overrun the homing sensor. After setting this system, it is recommended to test feed the system at least 6 times to make sure that the homing sensor is activated at rest for each position on the wheel.
- **Sort Motor Speed** – The speed of the sorter motor. Again, a value of around 90 is recommended for smooth operation. If your sorter is losing position, you can try reducing this value as an alternative to increasing voltage on the motor controllers.
- **Slot Step Inc** – Default value is 20. This represents the number of steps between each slot/tray/bucket on your sorter machine. The stepper motors for this project have 200 steps per revolution or 1.8 degrees per step. If you wanted to create more slots, you simply need to have a longer sorter tube outfeed. This will allow you to reduce the number of steps between slots. You would also need to increase the number of slots in the configuration screen.
- **Sort Homing Offset** – This is the number of additional steps the motor should turn after the homing switch has been triggered. It can help with centering your sort tube over the first slot (slot 0) but you must be careful not to use too large a value here as it can cause the system to overrun the homing sensor. Recommended range is between 0 – 8.
- **Motor Standby (sec)** – This represents the time in seconds to wait after motor inactivity to turn the motors off. Turning the motors off when not in use will help reduce heat and power consumption. It is generally recommended that this value be a number above 30 seconds, however lower values will work too. When the motor goes into standby, you may notice that the feed wheel reverses backwards a few millimeters depending on the strength of the tension cam spring. In the CS7.2 release, this setting is less necessary as the motors are “throttled down” to about 20% power which holds them in place without generating excessive heat.
- **Slot Drop Delay** – This represents the amount of delay (in milliseconds) the sorter will wait before moving to the next slot. This allows time for the brass to fall and clear the sorter tube before it moves to the next slot. You can decrease this time and improve performance if running on a slower computer as there will already be natural delay built in.
- **Notification Delay** – This is the delay (in milliseconds) that the Arduino will wait after a feed cycle has completed before sending the done command. Generally, the brass case is still moving slightly when the feed cycle completes, and this results in a blurry image. Adding a bit of delay here allows the resonance and movement to stop before the software captures an image. If you are getting motion blur in your images, you should try increasing this setting in increments of 20 to find the optimal/minimal delay.

Debounce: Debounce is a feature which helps to prevent the system from jamming when brass falls down the feed tube and activates the proximity sensor but then bounces back up while the wheel is turning. This is common if you don't have a backlog of brass in the feed tube or the collator/brass feeder is struggling to keep up.

- **Debounce Timeout** – The duration (in milliseconds) that must pass without the proximity sensor being triggered before the system schedules a debounce pause.
- **Debounce Pause Time** – The delay (in milliseconds) after the proximity sensor is triggered before the feed wheel cycle begins.
- **Feed Motor Current** – For CS7.2 systems and others that support current control via UART, this setting defines the maximum current (in milliamps) allowed for the feed motor.
- **Sort Motor Current** – For CS7.2 systems and others that support current control via UART, this setting defines the maximum current (in milliamps) allowed for the sorter motor.

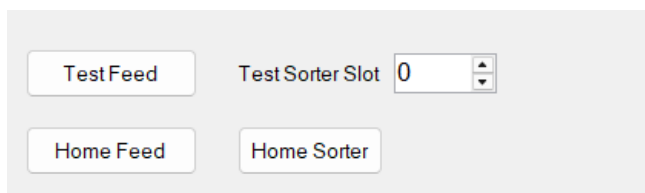
Air Drop Configuration - To read more about the airdrop mod, you can visit the Mods folder in the CS7.1 GitHub repo at: <https://github.com/sjseth/AI-Case-Sorter-CS7.1/tree/main/Mods/AirDrop>

- **Pre Drop Delay** – The duration in milliseconds (ms) after a feed cycle completes to wait before sending the signal to the AirDrop relay.
- **Signal Duration** – The duration (ms) of the signal sent to the relay. This controls how long the air blast is.
- **Post Drop Delay** – The duration (ms) after the signal to wait before sending the “ok” command back to signal the feed is complete. When enabled it takes the place of **Notification Delay**. This should typically be a value between 50-100. If some of your pictures have motion blur or the brass is not clearing the tube before the next feed, increase this value by 50 until the issue goes away.

Changes to the settings above will not be applied until the **Save Settings** button is pressed.

Testing

The configuration screen also has a testing section as seen in the screenshot below:



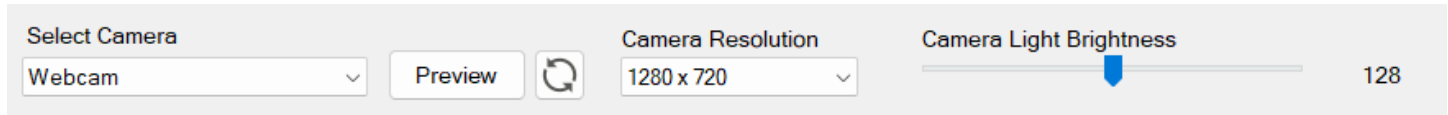
Most of the controls here are somewhat obvious but here is a summary of each:

- **Test Feed** – Will cause the feed wheel to move forward one position. Ignores the state of the feed proximity sensor and does a force feed (xf:0) command.
- **Test Sorter Slot** – Moves the sorter arm to the selected slot position by sending the force move command (sortto:1)
- **Home Feed** – This sends a homing command and causes the feed wheel to move to the next homing position if not currently homed.
- **Home Sorter** – This causes the sorter arm to move to its home position.
- **Slot Count** – This is not actually a “test” feature but a part of the software configuration and enabled the software to recognize more slots than the default of 8. **Note that if you increase the slot count, you will need to reduce the **Slot Step Inc** setting as there are only so many degrees in a circle. For more*

information and an example on how to use this feature, check out the Long Arm Hex Sort mod at: <https://github.com/sjseth/AI-Case-Sorter-CS7.1/tree/main/Mods/Long%20Arm%20Hex%20Sort>

Image Processing

Camera Selection and Resolution

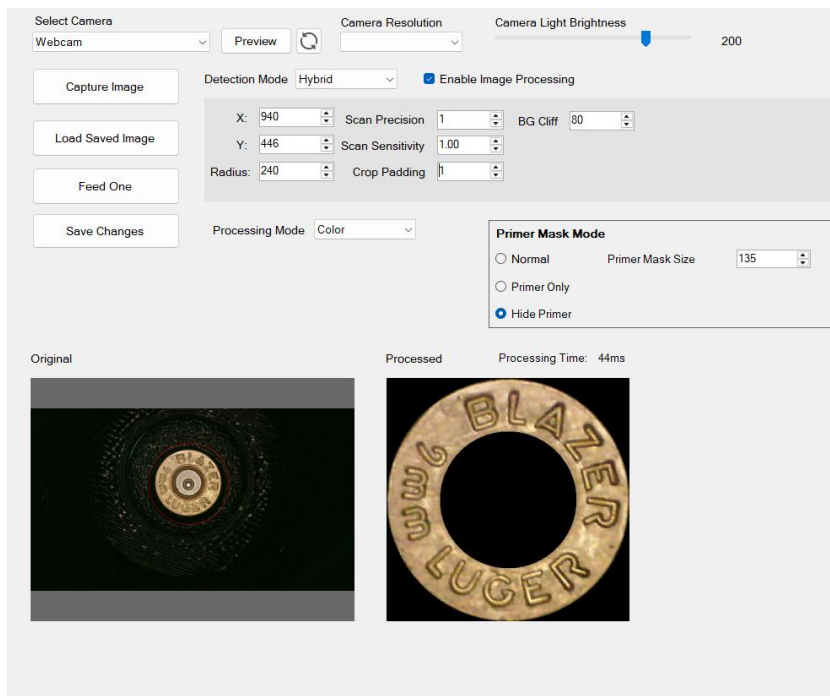


A list of connected web cameras will be displayed in the top corner of this screen. Clicking the Preview button will display a feed from that camera in the “Original” image preview box below.

There are two camera versions for this sorter build (v1 and v2). The recommended resolution for the v1 camera is 640x480 and the v2 camera is 1920x1080 or 1280x720. Select the appropriate resolution for the camera you have selected. For the CS7.2 build and beyond, we are only using the v2 camera.

Image Processing & Tuning

Image processing is the part of the software which is responsible (if enabled) for detecting, centering, and cropping the pictures of brass as well as creating rotations of those images. If this feature is not enabled, the raw image as captured by the camera will be used for the training.



The image processing is used to adjust the image processing settings used to detect, center, crop, mask the images as well as control color processing modes.

Some of these settings are stored in the currently active model and one should note that whatever settings you use here to capture and process images for your model, the same settings should be used when using the model in Run mode as a mismatch here will create poor recognition performance issues.

To begin tuning, you first need to load an image. You can use one of the 3 methods (buttons) on the left of the screen to do this. Here is how each of these methods works:

- **Capture Image** – This will capture an image from the Camera. If in emulation mode, it will randomly select an image from the emulation images folder.
- **Load Saved Image** – Use this feature to load a previously saved image. It defaults to the image processing tuning folder at: C:\Program Files\SJSeth\AI Brass Sorter\training\images\tuning. This is where images are saved when the **Save for IP Tuning** button is pressed in the **Training** screen.
- **Feed** – Similar to the functionality in the training screen, this initiates a feed cycle and then captures an image.

Detection Mode

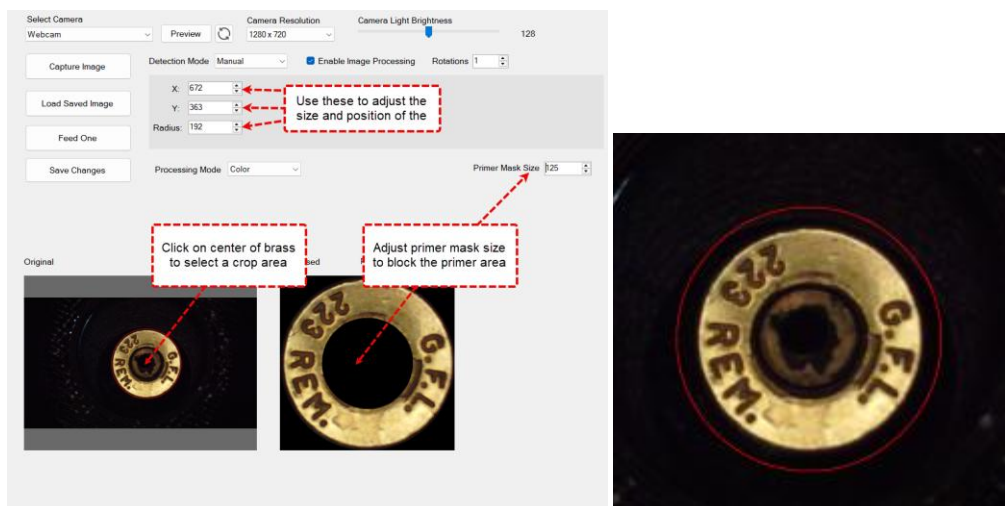
There are currently 3 detection modes which are used to locate, crop, and center the brass.

- **Manual** – User decides the crop area by selecting it. This is the fastest mode but often can create inconsistencies in the images as the brass can move a little from feed to feed.
- **Auto** – Machine detects the crop area automatically. This mode scans the entire image and can be a bit slow on some computers.
- **Hybrid** – (RECOMMENDED) Combination of Manual and hybrid where user selects a larger crop area and then uses automatic detection within the user's selection. This provides a balance of consistency and performance.

The following is a description of the settings for each of the above modes.

Manual Detection Mode

You can click on the original image to set the initial crop area (red circle around brass).



Radius

This will increase or decrease the size of the crop area circle.

X & Y

These will move the crop circle in 1 pixel increments along the X and Y axis.

Automatic Detection Mode

Detection Mode Automatic ▾

Scan Precision 1 ▴ ▾ BG Cliff 30 ▴ ▾

Scan Sensitivity 1.00 ▴ ▾

Crop Padding 1 ▴ ▾

Scan Precision

As the software scans an image, it scans both horizontal and vertical lines. The precision of 1 represents a scan of every line of the picture. Increasing this number causes a coarser scan of the image. For example, if a precision of 4 is selected, the software will scan every 4th line (4 pixels apart). This can provide some performance increases while still offering reasonable detection.

Scan Sensitivity

This setting is the amount of light change required to trigger a detection point or edge (edge of the piece of brass). We scan each line and create an average value for the pixels in that line. It is compared against the previous averages and if it rises above the defined sensitivity value, that line will be marked as an edge. The scanner does its scan from left to right, right to left, top to bottom, bottom to top. 0 represents black and 255 is white.

Crop Padding

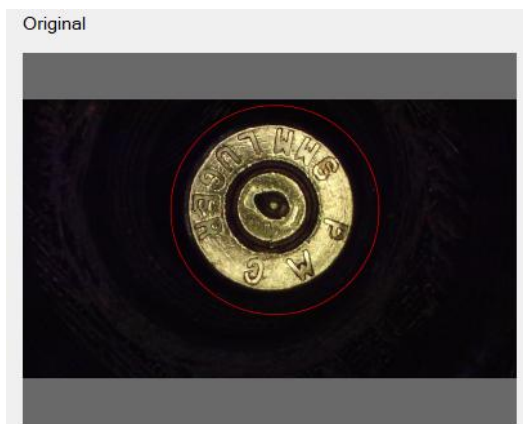
After we have detected the edges of the brass piece, we calculate the center and crop the image. The padding represents how much space to include outside of our crop or the black space which will be around our brass in the final image. It is recommended to have the least amount of padding possible.

BG Cliff

Background threshold determines the threshold at which a pixel is considered to be black or 0. This can be helpful to resolve issues where there is too much light or a gray background instead of a black background. A setting of 25 to 30 is generally recommended here.

Hybrid Detection Mode (Recommended)

Hybrid Detection mode simply combines the controls and functionality of Manual and Automatic mode. It is recommended that you select a large enough manual crop area to accommodate the positional changes from feed to feed. It is a good idea to set the area and then hit the feed button several times and pay attention to see that the brass is well within the crop area for each. See example below.

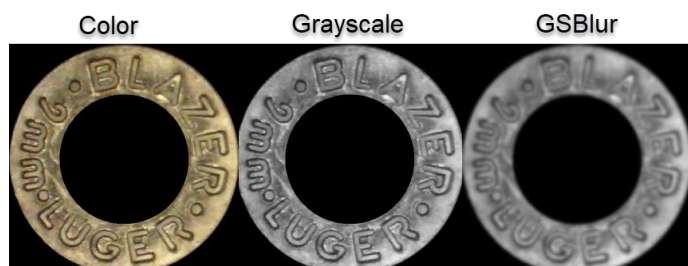


Processing Modes

The processing mode selection is used to provide additional image processing to the cropped images. Here is a summary of each mode.

- **Color (Recommended)** – This is the default mode and applies no color changes to the original image. In this version of the software, when training the ConvNeXt models, we are automatically generating rotations and desaturations and other variations so starting with the most information possible is the best idea here.
- **Grayscale** – Desaturates the image so that it is rendered in grayscale.
- **GSBlur** - A combination of grayscale and blur

Here is a visual comparison of the different modes listed above.



Processing Mode Configurations

GSBlur

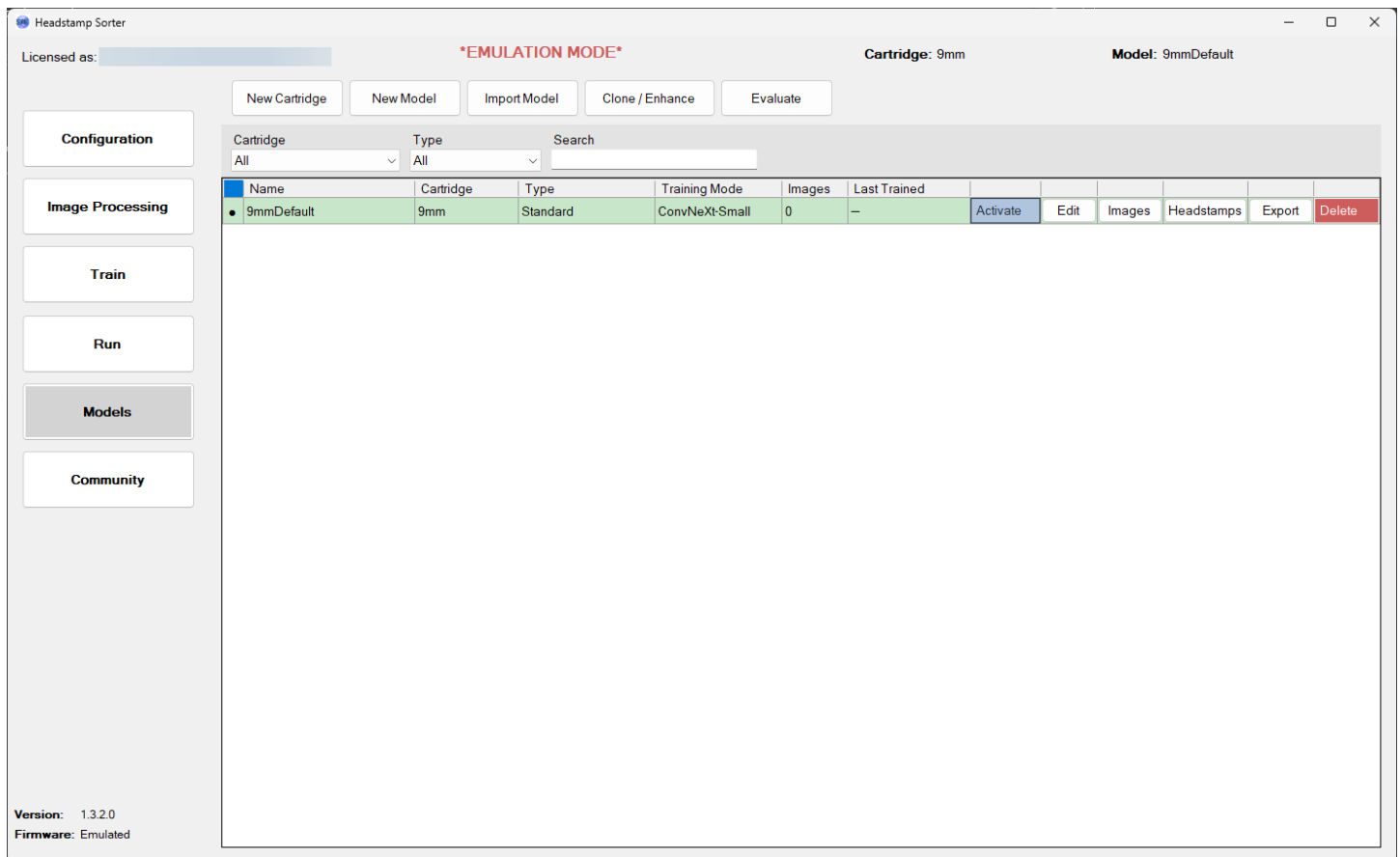
Blur Size – The area used for calculating blur. Generally, a range between 2 and 40.

Blur – The amount of blur to be applied. (recommended range of 1-3)

Models Management

The **Models** screen is where you go to manage your cartridges and models. A cartridge is simply a high-level classification in which to organize subsequent models. For instance, by default there is cartridge named “9mm”. Within the 9mm cartridge, you can have many different AI Sorting models for different types of sorting of 9mm brass. You can create as many models as you need within a cartridge.

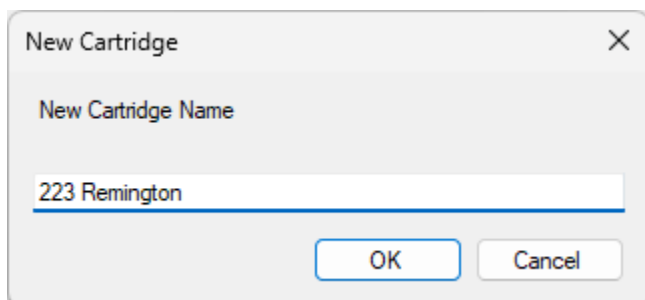
A model is a collection of images with headstamp associations (classifications) which are used to train a machine learning neural network. Every time a training is performed, a snapshot zip file is created which represents the trained neural network. If the file already exists, it will be overwritten with the new trained model. If you add, remove, or reclassify images in the training screen, you will need to rerun the training to regenerate the model file.



Creating, Editing and Deleting Cartridges

To add a new cartridge, you need to hit the **New Cartridge** button at the top of the screen

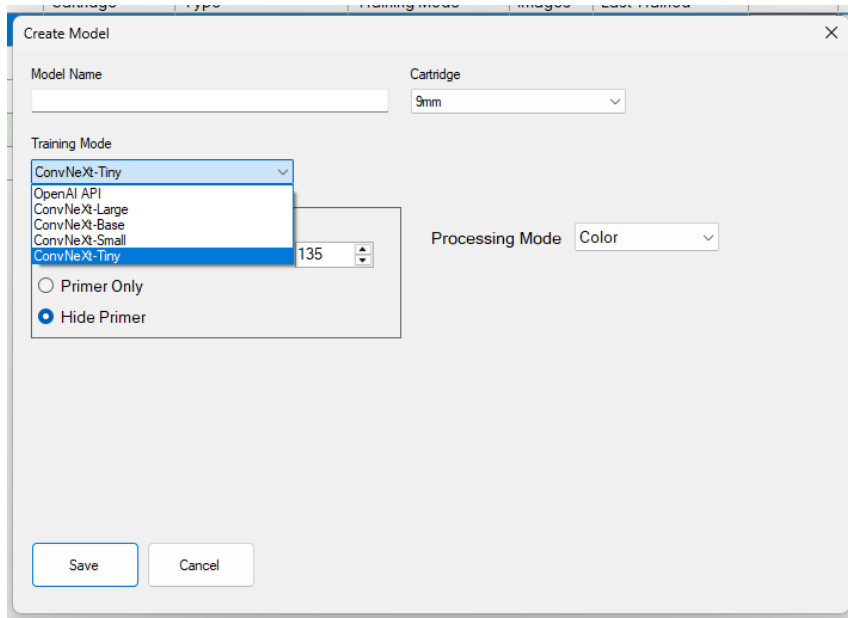
Give the new cartridge a name and hit the OK button.



When you create a new cartridge a default model is also generated for that cartridge

Creating and Editing Models

To create a new model, click on the New Model button at the top of the screen. This will display a creation screen like the screenshot below.



The first task in creating a model would be to give the model a name. The model's name should be short and concise and you should avoid special characters such as @#%*&()/, etc. Here are some examples ("45_binary_largePrimer", "9mm Nickle", "223_556")

The name should provide enough information to identify the model.

Model Type (normal, mask, primer-only)

There are 3 radio buttons to select the type of model masking that is applied to your model.

- **Normal** – No masking applied.
- **Primer Only** – Uses the Primer Mask Size feature along with Image processing to crop the defined primer area. This mode is used to “zoom in” on the primer area and ignore the rest of the case headstamp which is useful for separating large and small primer brass which is common in some cartridges such as 45 Auto. *This requires image processing to be enabled in the configuration screen.
- **Hide Primer (Recommended)** – This is recommended for most models as it helps to improve the recognition of the headstamps by ignoring the primer area altogether. When this is enabled, a mask (black dot) will be applied to the primer area. *This requires image processing to be enabled in the configuration screen. Changes made to this setting will only be applied to newly captured images and will result in a mismatch for newly captured images. To apply this setting to already captured images, you should use the **Model Enhancer**.

Here is a representation of a resulting image from the 3 different types:

Normal

Hide Primer

Primer Only



Primer Mask Size

This defines the size of the primer mask. For hiding a primer, generally a value of 135 works well here. For primer only, you may want to increase the size to 145-150. This value will also update the mask size value in the Image Processing **Fine Tuning** screen. Changes to this value in that screen will be applied to the model as well. It is important to know that changing this setting after you have already captured images can be problematic as this setting only applies during the capture process and newly captured images will not match those which were previously captured. To apply this setting to already captured images, you should use the **Model Enhancer**.

Processing Modes

The processing modes were covered in detail earlier in this guide in the Image Processing section. Please refer to that for more information. If using a mode other than “Color”, image processing needs to be enabled. Changes to this value in that screen will be applied to the model as well. It is important to know that changing this setting after you have already captured images can be problematic as this setting only applies during the capture process and newly captured images will not match those which were previously captured. To apply this setting to already captured images, you should use the **Model Enhancer**.

Training Mode

The software supports all the older modes for backwards compatibility however when creating new models, only the ConvNeXt modes are available. These modes affect the size and training time of the model. For typical models, Tiny and Small are more than adequate to achieve accuracy results above 99%. The recommendation is to use Tiny. You can change this setting later if you want to experiment with other modes.

Editing a Model

To edit an existing model, click the **Edit** button in the row of the model you wish to edit.

Cartridge

Type

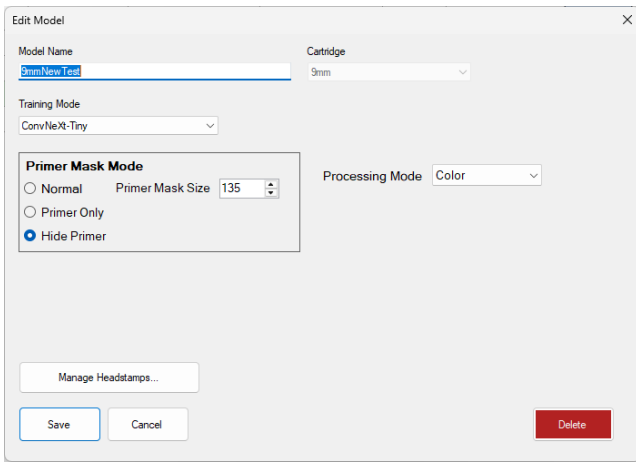
Search

All

All

Name	Cartridge	Type	Training Mode	Images	Last Trained	Activate	Edit	Images	Headstamps	Export	Delete
• 223 RemingtonDefault	223 Remington	Standard	ConvNeXt-Tiny	0	—	Activate	Edit	Images	Headstamps	Export	Delete
9mmDefault	9mm	Standard	ConvNeXt-Small	0	—	Activate	Edit	Images	Headstamps	Export	Delete

The **Edit** screen looks very similar to the model creation screen but has one additional button to launch the headstamp manager which allows you to create, rename or remove headstamps which have been added to the model as well as manage parent relationships. This feature will be covered in detail later in this document and is also available directly in the models table by clicking the “Headstamps” button.



Model Name:

Cartridge: 9mm

Training Mode: ConvNeXt-Tiny

Primer Mask Mode:

- ☐ Normal
- ☐ Primer Only
- ☒ Hide Primer

Primer Mask Size: 135

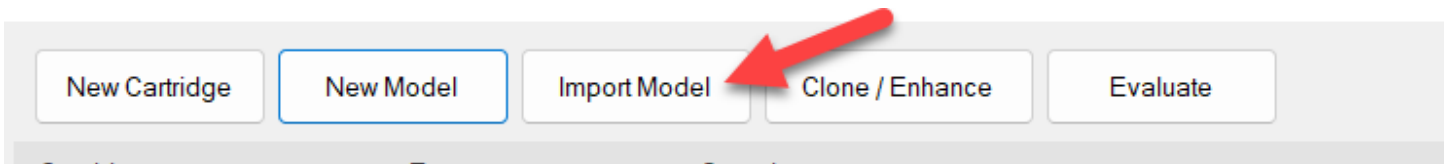
Processing Mode: Color

Manage Headstamps...

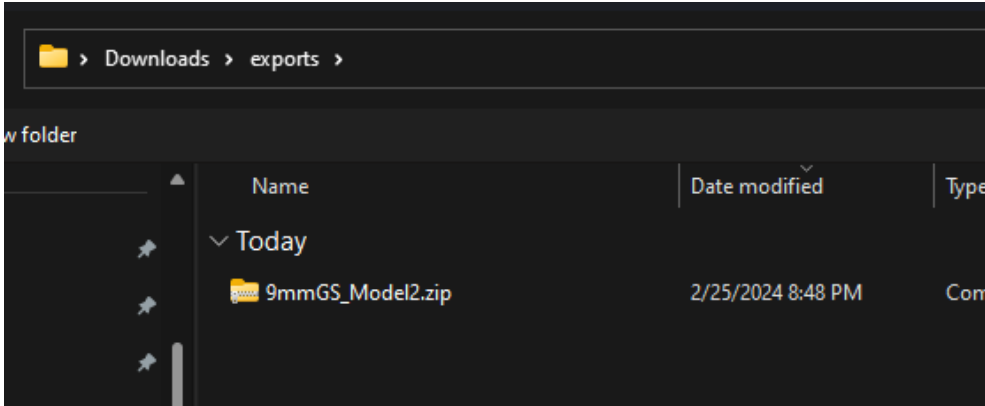
Save Cancel Delete

Import

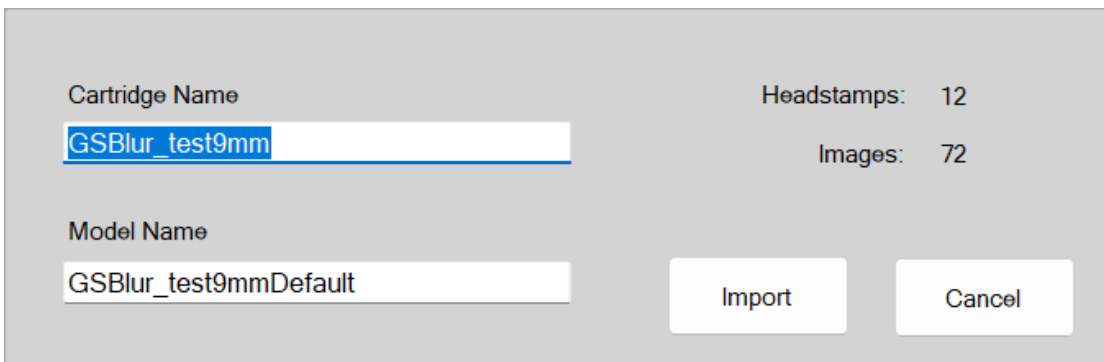
The import feature can be used to import an exported or shared model. The model can be one which was created on a different system and exported to an export file.



After hitting the **Import** button, a file dialog will open, and you can browse to the export file which you intend to import.



Once you have selected the file you wish to import, you will see a summary screen where you can change the name of the cartridge and model to be imported. If the cartridge name already exists, the model will be associated with it otherwise a new cartridge will be created as well.



Cartridge Name:

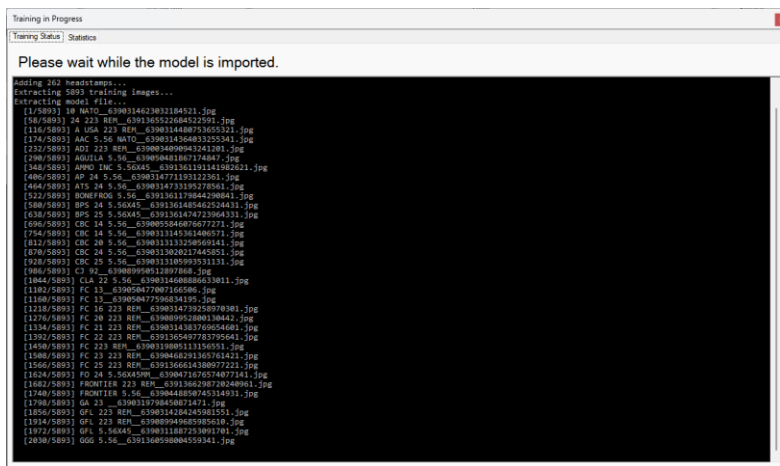
Headstamps: 12

Images: 72

Model Name:

Import Cancel

While the model is importing, you will see a status screen of the progress:



Export

The export feature is used to create an importable model file containing the model information and optionally the model and the training images. The model and images are compressed into a zip file along with a manifest file which contains information about the model. That zip file can then be shared or saved as a backup of that model. To export a model, simply click the export button in the row of the model you wish to export.

Cartridge	Type	Search								
All	All									
Name	Cartridge	Type	Training Mode	Images	Last Trained					
223 RemingtonDefault	223 Remington	Standard	ConvNeXt-Tiny	0	—	Activate	Edit	Images	Headstamps	Export
9mmDefault	9mm	Standard	ConvNeXt-Small	0	—	Activate	Edit	Images	Headstamps	Export

Once the export is selected, you will see a dialog which lets you configure the export and select whether or not to include the images or trained model.

Export Model Options

☐ Include Training Images

☐ Include Trained Model

OK

Cancel

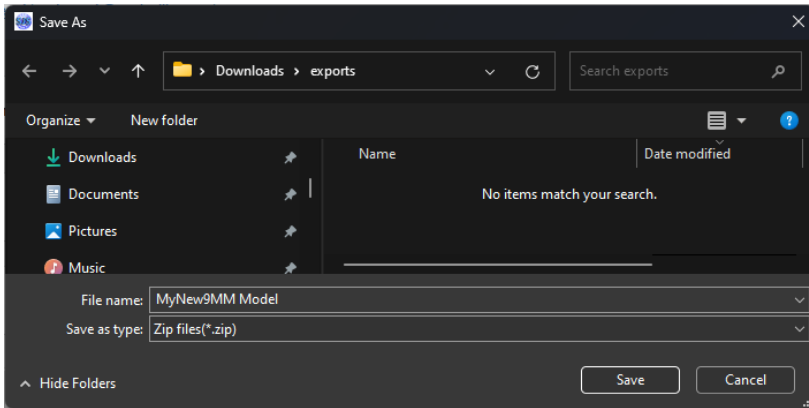
I am sure the question would come up as to why one would want to include or omit images or models so I will briefly cover a few scenarios.

No Boxes Checked - I only want to share my headstamp names and parent relationships. This can serve as a guideline. Also for OpenAI API models, there are no “snapshot models” or images so this only shares the configuration (prompts). (Passwords/keys are not exported)

Only Training Images – If you just want to zip up the images in a model to keep as a backup or share with others to let them create their own headstamp names, this can be useful.

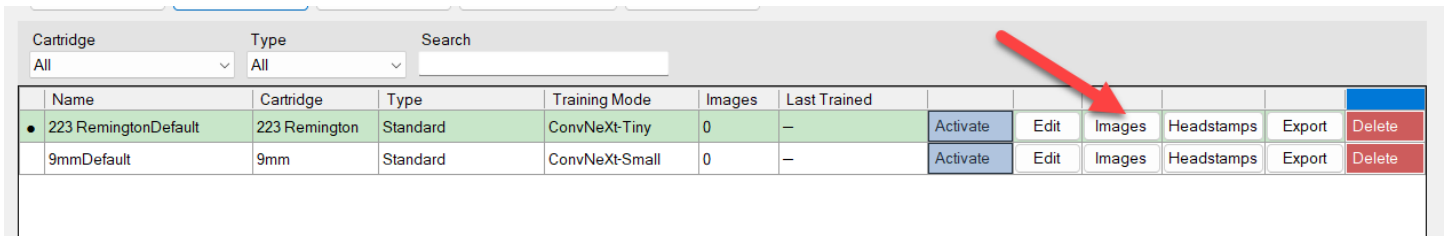
Only Models – If you are want to import into a system that is only going to be doing classification and not training models, then you only need to include the model. This significantly reduces the size of the export.

Once you have selected the export options and click the OK button, a file dialog will open which allows you to select the location and name of the exported file.



Managing Images

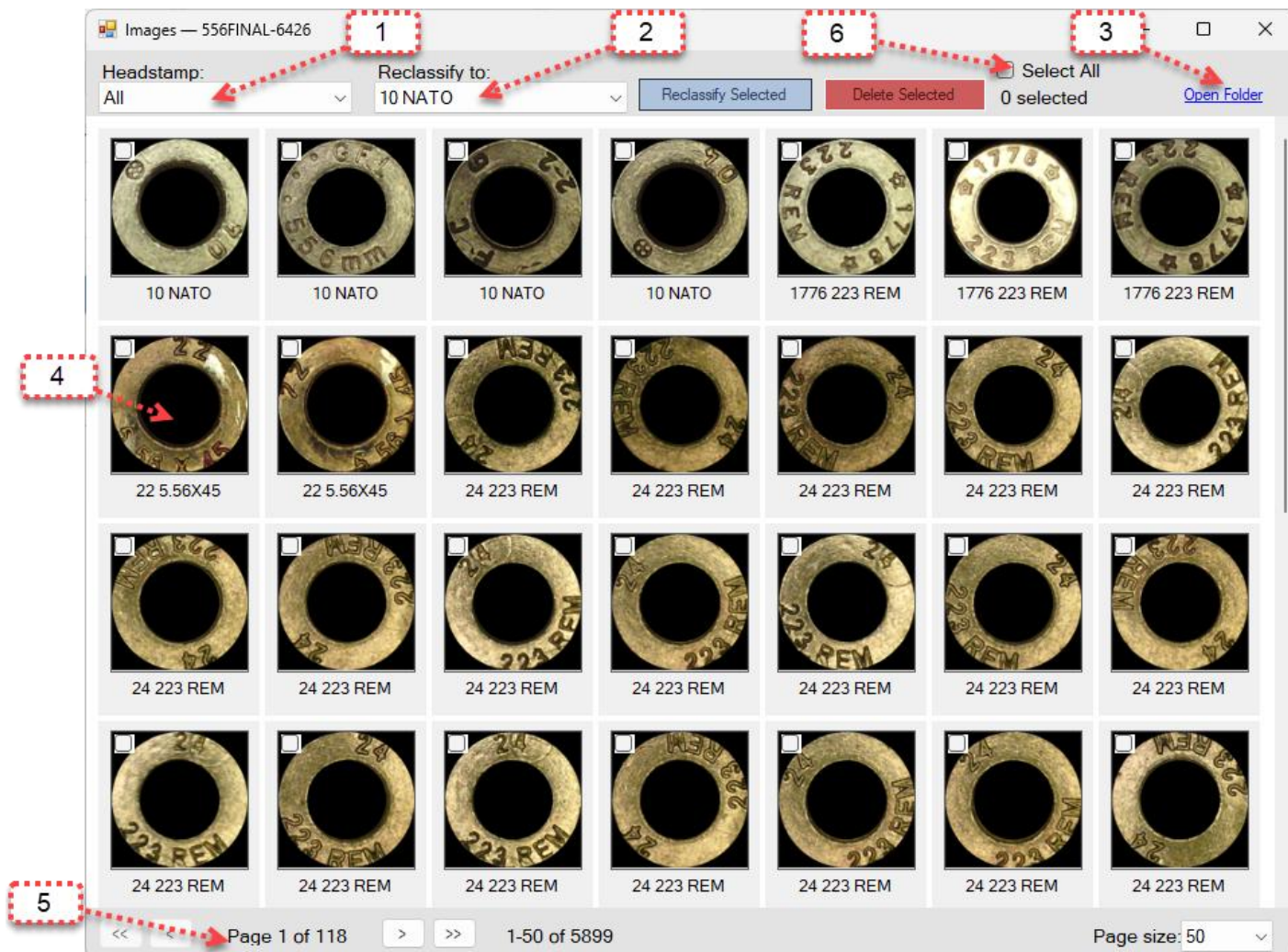
The **Images** screen is where you can go to inspect the training images for your model. This screen also gives the capability to change the classification for an image if one was misclassified.



Name	Cartridge	Type	Training Mode	Images	Last Trained	Activate	Edit	Images	Headstamps	Export	Delete
223 RemingtonDefault	223 Remington	Standard	ConvNeXt-Tiny	0	—	Activate	Edit	Images	Headstamps	Export	Delete
9mmDefault	9mm	Standard	ConvNeXt-Small	0	—	Activate	Edit	Images	Headstamps	Export	Delete

The following numbered items correspond to the numbered labels in the screenshot below outlining the different functions of the **Manage Images** screen.

1. Dropdown menu to filter the display by a particular headstamp
2. A dropdown menu which lets you reclassify all the selected (checked) images in one step
3. The **Open Folder** link opens the training image file location using the operating system dialog. This can be helpful if you want to do bulk operations manually.
4. Clicking an image brings up a dialog to let you reclassify or delete that image.
5. Grid paging controls. With a page size of 40 items, these arrows allow you to page through the images.
6. Select All visible items. (selects only those items visible on the current page)



You can also click on an individual image to see the image in larger format with the option to reclassify or delete.



Managing Headstamps

In the model management screen click on the “Headstamps” button in a row to manage the headstamps for a model. This utility allows you to create, edit and delete headstamps in the model and also allows you to define “parent” headstamp classifications which can be useful for certain run operations.

Name	Cartridge	Type	Activate	Edit	Images	Headstamps	Export	Delete
223 RemingtonDefault	223 Remington	Standard	Activate	Edit	Images	Headstamps	Export	Delete
45AUTOMaster	45 AUTO	Standard	Activate	Edit	Images	Headstamps	Export	Delete
45AUTOMaster_PM	45 AUTO	Standard	Activate	Edit	Images	Headstamps	Export	Delete
556FINAL-6426	556	Community (trainable)	Activate	Edit	Images	Headstamps	Export	Delete

Headstamp Manager — 556FINAL-6426

Parents

- ADI
- AGUILA
- AP
- BPS
- CBC
- CJ
- FC
- FO
- FRONTIER
- GA
- GFL
- GGG
- HORNADY
- IK
- IMI
- IVI
- JAG
- L
- LC
- M
- MKE
- MON
- NORMA

Headstamps

Name	Parent		
CBC 22 5.56	CBC	Edit	Del
CBC 23 5.56	CBC	Edit	Del
CBC 24 5.56	CBC	Edit	Del
CBC 25 5.56	CBC	Edit	Del
CJ 92	CJ	Edit	Del
CJ 93	CJ	Edit	Del
CLA 22 5.56	(none)	Edit	Del
FC 06	FC	Edit	Del
FC 07	FC	Edit	Del
FC 10 223 REM	FC	Edit	Del
FC 11 223 REM	FC	Edit	Del
FC 12	FC	Edit	Del
FC 12 223 REM	FC	Edit	Del
FC 13	FC	Edit	Del
FC 13 223 REM	FC	Edit	Del

Buttons: Add Parent, Rename, Delete, Auto-Suggest Parents, Add Headstamp, Save

The following numbered items correspond to the numbered labels in the screenshot above outlining the different functions of the **Headstamp Manager** screen.

- 1. Parent Headstamps List** – A list of defined parent headstamps
- 2. Headstamp Grid** - The dropdown menu in this grid let's you map a headstamp to a parent headstamp.
- 3. Edit and Delete headstamps** - Using the **Edit** button in a row allows you to change or rename a headstamp. This also renames all of the corresponding images in the training set to the new headstamp name. You will need to retrain your model if you modify a headstamp's name or **delete** a headstamp
- 4. Add Parent** – Allows you to add new parent definitions
- 5. Auto-Suggest Parents** – This feature creates and auto assigns parent classifications based on the names of existing headstamps. Generally it looks at the first part of the headstamp name and if there are multiple headstamps which share the first matching segment (eg: **FC** 07, **FC** 08), it creates a parent classification and assigns it to all headstamps with that matching segment.

6. **Add Headstamp** – This adds a new headstamp classification to the model.
7. **Save** – This button saves the parent headstamp relationships. Other operations on this page are performed in real-time.

Parents, what are they good for?

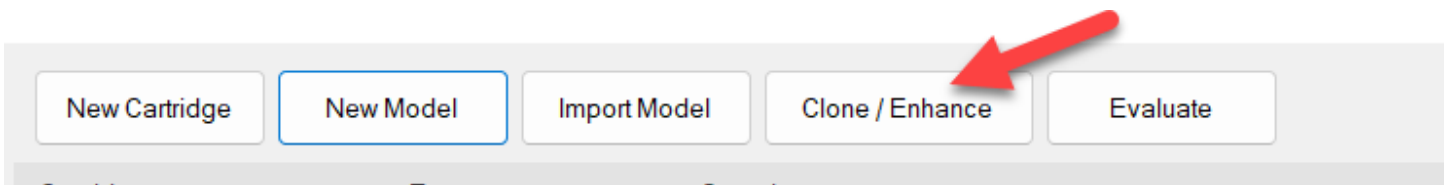
Parent classifications can have multiple applications in sorting. When parents are defined in a model, these associations are available in the run screen which let us quickly group headstamps by parents. The run screen also has an option to sort by the parents directly. This can be useful if you have a model like the one seen in the previous screenshot where you have many variants of a manufacturer's headstamp but want to group them all together.

Besides manufacturer grouping, here are some other parent classifications which could be useful for certain models:

- Nickle, brass, aluminum, steel
- Junk, Not Junk
- Small Primer, Large Primer

Model Clone / Enhance

You can access the model cloner and/or enhance by first selecting the model row and then clicking the **Clone / Enhance** button



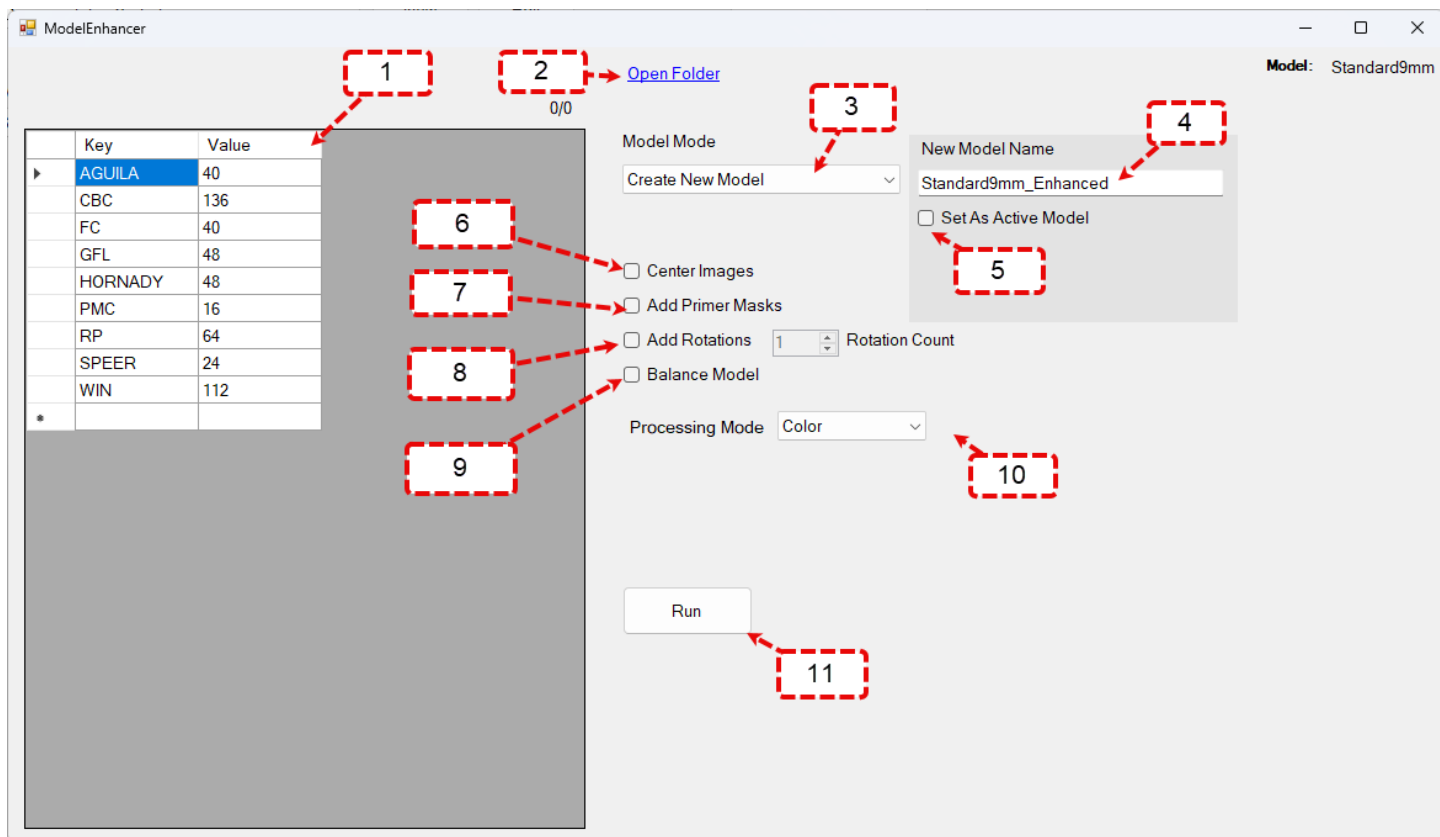
The Model Enhancer utility was created to be a multi-purpose tool with functions to “enhance” a model. The enhancements can be applied to an existing model or can be applied to a clone of the existing model thus creating a new model and leaving the existing model unchanged. It is highly recommended to create the new model whenever possible as it preserves the existing model and allows you to “roll back” or discard the new model if the changes ended up not working as well or being less than desired. Multiple models can be created based on a “base” model which allows for a/b type testing between the different models.

The following numbered items correspond to the numbered labels in the screenshot below outlining the different functions of the **Model Enhancer** screen.

1. The headstamp grid contains the headstamps and images counts for each that are present in the model.
2. The **Open Folder** link opens the images folder on the file system for the current model.
3. **Model Mode** menu selects whether the selected **Run** operations will be performed on the current model or to a new model which starts from a clone of the current model.
4. If “Create New Model” is selected in #3, the **New Model Name** field will be the name of the newly created model.
5. If selected the newly created model will be set as the current and active model in the system, else the currently selected model will remain active.
6. **Center Images** – Using automatic detection, if this option is selected each image in the training set will be processed and centered. This can be useful if you have excess padding around your images and want to add consistency. Because the processing here is automatic, it is recommended to visually inspect the resulting centered images in thumbnail view in the operating system images folder (see #2)

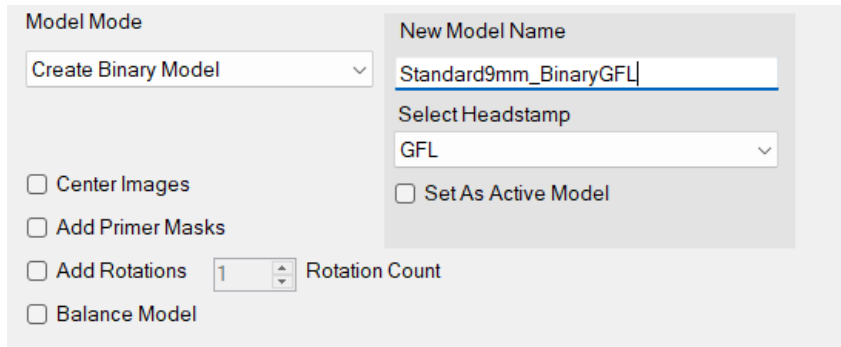
7. **Add Primer Mask** – If selected, a mask will be placed over the center of the image with the primer mask size defined in the selected model. Note the mask is a black circle that is blindly placed in the center of the image. If your images are not already centered, this will yield poor results.
8. **Add Rotations** – This function will add the specified number of rotations for each image in the model. Degree of rotation for each image is based on the 360 divided by the **Rotation Count**.
9. **Balance Model** – This feature is designed to ensure that each headstamp classification has the same number of images. In testing there seems to be a strong correlation between accuracy and having a balanced image set. It performs this balance operation by adding rotations to the images in each classification in the model until it has the same number of images as the largest classification set. In the screenshot below, CBC with a count of 136 would represent the largest set. In this example, roughly 2 rotations would be added to each image in the Aguila set bringing the number to 120 and then 1 additional rotation would be added to the first 16 images until the 136 number has been reached.
10. **Processing Mode** – This menu is the same control set that is seen in Model Create/Edit screen and the Fine Tuning screen. See that section of this document above for more details on the individual modes and settings. In this context, the images will be processed with whichever processing mode is selected. This can be useful if you want to change all your images to a new mode such as Grayscale. If you already have a good model but are seeking to improve it, you can clone the model and use one of these modes. This will allow you to compare the accuracy of your existing model vs the one newly created with the new color mode.
11. The **Run** button begins the model enhancement operations. The order of operations is as follows:
 1. If **Create a New Model** is selected, a new model will be created and then all of the images and headstamps will be cloned and associated to the new model.
 2. Image Centering (if selected)
 3. Add Primer Masks (if selected)
 4. Add Rotations (if selected)
 5. Balance Model (if selected)
 6. Apply Color mode (if different than base model)

The progress of the run operation will be seen in a progress bar which is above the headstamp grid.



Creating a Binary Model

The Model Enhancer also contains a feature to create a binary model based on an existing model.

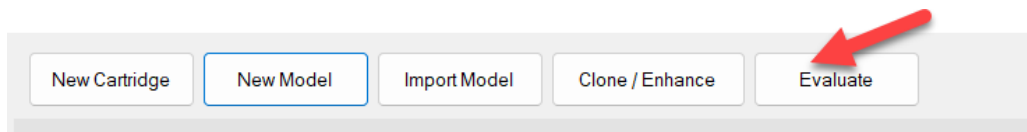


The screenshot shows a dialog box for creating a binary model. It has two main sections. The left section, titled 'Model Mode', contains a dropdown menu set to 'Create Binary Model' and four checkboxes: 'Center Images', 'Add Primer Masks', 'Add Rotations' (with a value of 1 and a 'Rotation Count' label), and 'Balance Model'. The right section, titled 'New Model Name', contains a text input field with 'Standard9mm_BinaryGFL', a 'Select Headstamp' dropdown menu set to 'GFL', and a checkbox labeled 'Set As Active Model'.

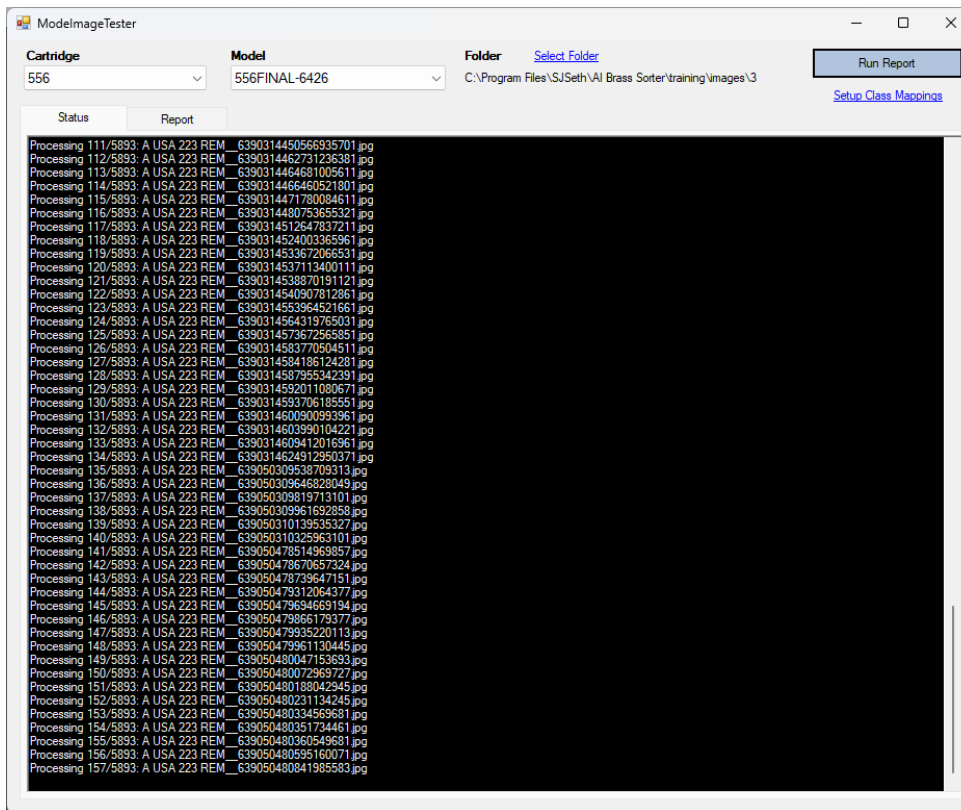
A binary model is a model with two classifications (EG: hotdog, not hotdog). When binary is selected as the mode, you will see an additional dialog which is a menu of all the classifications for that model. You must select the classification you wish to create the model based on. To create the model, the system will use the count of the currently selected headstamp images and select a balanced set of images from all the other classifications so that the total counts between the two classifications are equal. (GFL, not GFL)

Evaluating a Model

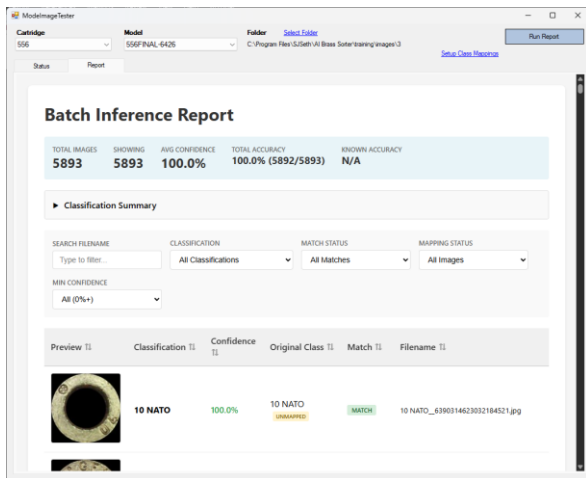
This is a new screen to v1.2x and newer. You can access model evaluation by clicking the Evaluate button



This launches the model evaluation screen which allows you to evaluate a model against its own training images or optionally select another folder of images to evaluate with the model. The evaluation is performed by comparing the model prediction against the actual classification listed in the file name. You can use the “**select folder**” link to point the evaluation at a different image folder than that of the model.



Once the evaluation is complete, you will see a report rendered in the Report tab. The report will give a high level overview of the model’s performance.



You can also expand the Classification summary to get a breakdown of how each headstamp performed in the evaluation.

▼ Classification Summary					
Classification	# of Images	Avg Confidence	High	Low	Mismatched
10 NATO	4	99.8%	100.0%	99.5%	0
1776 223 REM	3	99.9%	99.9%	99.9%	0
22 5.56X45	2	99.9%	99.9%	99.9%	0
24 223 REM	60	100.0%	100.0%	100.0%	0
85 13	5	100.0%	100.0%	100.0%	0
A USA 223 REM	94	100.0%	100.0%	100.0%	0
AAC 5.56 NATO	62	100.0%	100.0%	100.0%	0
AB 5.56	1	99.9%	99.9%	99.9%	0
ADI	3	99.9%	100.0%	99.7%	0
ADI 223 REM	18	100.0%	100.0%	100.0%	0

You can also use the filters above the evaluation grid to quickly find those images which have lower confidence or where the classification didn't match

SEARCH FILENAME CLASSIFICATION MATCH STATUS MAPPING STATUS MIN CONFIDENCE					
Type to filter... All Classifications Doesn't Match All Images All (0%+)					
Preview	Classification	Confidence	Original Class	Match	Filename
	MKE 14	98.4%	MKE 24 UNMAPPED	MISMATCH	MKE 24_6391361427443257111.jpg

This can be helpful to find those images in your model training folder which were misclassified.

The Train Screen

Training a Model (Train)

The process of training a model consists of a few basic steps:

- Capture an image (and optionally, its rotations)
- Classify the image (assign a headstamp)
- Save the image(s) to the model's images folder.
- Repeat the previous steps until enough images have been captured and classified. To perform the machine learning step, you need a minimum of two different classifications.
- Train the Neural Network (Train Model).

Controls

1 Feed 2 Add Image 3 Train Model 4 Feed Batch

Headstamp Classification: PMC 223 REM 7

Confidence: 95% 8

New Headstamp: 12

Sort While Training: 14 Skip Prediction: 23

Use Rotations: 15 Rotations: 6

Button Mode: [Add]→Feed [Save]→Add→Feed 16

Processed: 18

Raw: 19

Prediction: PMC 223 REM 20

Save for IP Tuning: 17

Images Folder: 10

Add Images on Click

Key	Value
10 NATO	4
1776 223 REM	3
22 5.56X45	2
24 223 REM	60
85 13	5
A USA 223 REM	94
AAC 5.56 NATO	62
AB 5.56	1
ADI	3
ADI 223 REM	18
AGUILA 223 REM	10
AGUILA 5.56	39
AMMO INC 5.56X45	60
AMTC	1
AP 16 5.56	2
AP 20 5.56	25
AP 22 5.56	2
AP 223 REM	2
AP 23 5.56	4
AP 24 5.56	13
ATS 24 5.56	64
BBR 223 REM	4
BHA 223 MATCH	1
BLANK	8
BLH 18 5.56	5
BONEFROG 5.56	60
BPS 24 5.56X45	60
BPS 25 5.56X45	59
CBC 08 5.56	1
CBC 09 5.56	22
CBC 14 5.56	60
CBC 16 5.56	1
CBC 17 5.56	2

11 11

Here is a brief description of each of the controls on the training screen and their function.

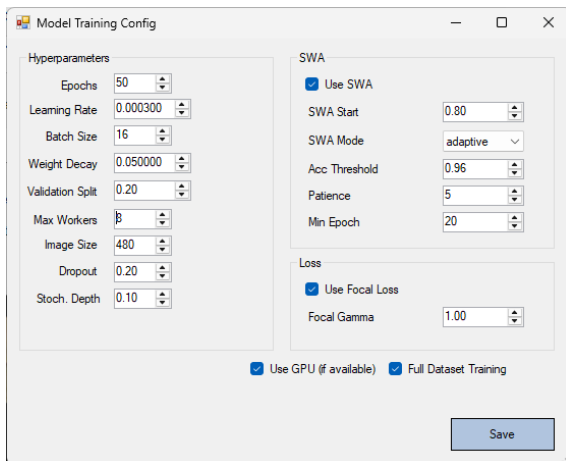
1. **Feed** – This causes the feed wheel to move one position. It ignores the status of the proximity sensor and performs a forced feed. Once the feed is complete, an image is captured and displayed in #19, #20, #11 (if use rotations #15 is enabled). Feed does not “save” images or modify the training set in any way (even with automations enabled). If you get a bad or empty image, use Feed to move to the next case.
2. **Add Image** – This button “saves” the images to the training folder with the selected headstamp classification (#7 and/or #10)

3. **Train Model** – This button initiates a training sequence where the images are processed through the neural network and a machine learning model is created. This can take anywhere from a few minutes to many hours depending on the number of images in the model’s training folder.
4. **Feed Batch** – Batch feeding is an automated option for adding many images of the same classification to a model quickly. For instance, if you have 50 pieces of brass with a classification of CBC and want to automatically add them, you will clear your feed system and then add those 50 pieces in so that only those 50 will be fed. Then you would change the classification value (#7) to CBC or (if CBC doesn’t exist yet, create it using #12). Once the classification has been set, you can press this button to begin automatically feeding and saving the images. The speed control for this function is #9.
5. **Images Folder** – This link opens the folder which contains your training images.
6. **Advanced Settings** – This link opens the advanced training settings dialog which allows you to select the settings to use while training your model. This is covered in detail later in this guide.
7. **Headstamp Classification Menu** – This is a list of all headstamps in your model. After an image is captured for a new case, you will use this menu (or optionally the grid in #10) to classify the image (and its rotations) before adding them to the model using the Add button (#2). *Note: after a model has been trained using the **Train Model** button, each time an image is captured, the software will attempt to automatically classify the image based on its training. It will automatically select the headstamp from this menu but will often be incorrect early in the training cycles but will improve as you add more images and retrain the model.
8. **Confidence** – This is often misunderstood as accuracy, and it is important to have a proper perspective on this value. It is a percentage-based confidence level given to a prediction based on what the model has been trained on. It can be highly confident and incorrect at the same time. For example, if you only added one classification to a model and trained it (not currently possible but just an example), because it only knows of one classification, it will be highly confident that every piece of brass it predicts will be that one classification. If you only have two classifications, it would be highly confident that it has selected the best match out of the two options. The number of unique headstamps in a model will also affect the confidence level but not the accuracy of. For instance, if you have 100 different headstamps in a model, and it predicts for 1 case, it is very common for it to predict correctly with low confidence. In this example if it had a confidence of 40%, it means that the 99 headstamps share the other 60%. You may end up with a confidence ranking like: Fed: 60%, CBC: 16%, Win: 3%, FC: 1%, LC: 1%, etc.
9. **Batch Classification Speed Control** – This slider determines how quickly to feed the batch when using batch classification.
10. **Headstamp Classification Grid** – This table gives a quick view of the headstamps in the model and how many images exist for each. It can also be used to classify images as an alternative to menu in #7.
11. **Image Rotations Preview** – If image rotations are enabled, this panel provides a preview of the rotations.
12. **New Headstamp Field** – As you are training your model, when you encounter a new headstamp, you can add it to your headstamp list by typing the headstamp (ensure unique) and hitting the **Save** button (#13). This will add it and select it in the headstamp menu #7.
13. **Save Headstamp** – Adds the headstamp classification option specified in #12 to the model (see #12)

- 14. Sort While Training** - If enabled, cases will be sorted based on their classifications into the bins specified in the **Run** screen configuration.
- 15. Use Rotations** – If checked, when a new image is captured, a number of rotations will be created. That number is based on the value set in the Image Processing section of the **Configuration** screen. *Note: This option will only be available if **Enable Image Processing** is checked in the **Configuration** Screen
- 16. Button Mode** – This menu lists a set of automation configurations for the buttons on the training screen. Here is a brief description of each:
- Standard** – No Automation, each button must be pressed to perform its action.
 - [Add]->Feed** – After the **Add** button is pressed and images have been added, a **Feed** occurs automatically (as if the feed button was pressed)
 - [Save]->Add** – When a new headstamp is added and the Save button is pressed, it will classify the image with that headstamp and add it to the images folder as if the **Add** button was pressed.
 - [Save] ->Add->Feed** – In addition to the mode above, it will also perform a **Feed** operation.
 - [Add]->Feed | [Save]->Add-> Feed** – This activates the above-described modifications for both the **Add** and the **Save** buttons and is the recommended mode as it is the most convenient.
- 17. Save for IP Tuning** – This button will save the original (unprocessed) image to the Image Tuning folder (typically at: C:\Program Files\SJSeth\AI Brass Sorter\training\images\tuning). Images saved to this folder can be loaded in the Image Processing Fine Tuning screen. (see [Image Processing](#))
- 18. Processed Image Preview** – This image is a preview of the fully processed image. This can be used to get a better view of the headstamp so that you can classify or reclassify the image.
- 19. Raw Image Preview** – This image is the unprocessed image as detected by the camera.
- 20. Headstamp Prediction** – Once the model has been trained, it will automatically try to predict each captured image. You can compare this value against the actual image to ensure they match before adding the image to the training set.
- 21. UNDO** – This will remove the last added image set from the images folder. If you accidentally misclassify and add the image, this will UNDO that mistake.
- 22. Add Images on Click** – This checkbox “enables” the headstamp grid to allow it to be used for classification. When unchecked, the classification grid has no clickable function.
- 23. Skip Prediction** – This checkbox signals that you don’t want the model to try to predict the headstamp during each feed in the training screen.

Advanced Training Settings

This screen allows you to customize/configure the settings which will be used to train your model.



The screenshot shows a 'Model Training Config' window with the following settings:

- Hyperparameters:**
 - Epochs: 50
 - Learning Rate: 0.000300
 - Batch Size: 16
 - Weight Decay: 0.050000
 - Validation Split: 0.20
 - Max Workers: 8
 - Image Size: 480
 - Dropout: 0.20
 - Stoch. Depth: 0.10
- SWA:**
 - ☒ Use SWA
 - SWA Start: 0.80
 - SWA Mode: adaptive
 - Acc Threshold: 0.96
 - Patience: 5
 - Min Epoch: 20
- Loss:**
 - ☒ Use Focal Loss
 - Focal Gamma: 1.00
- ☒ Use GPU (if available)
- ☒ Full Dataset Training
- Save** button

Here is a detailed description of what each setting does:

Epochs – The total number of times the model trains over the full dataset. Higher values give the model more opportunity to learn but increase training time. Recommended to have a minimum of 10 epochs.

Learning Rate – Controls how large each weight update step is during training. Lower values train more slowly but more precisely; too high and the model may overshoot and fail to converge.

Batch Size – The number of images processed together in a single training step. Larger batches give more stable gradient estimates; smaller batches use less memory and can generalize better.

Weight Decay – A regularization penalty applied to large weights to help prevent the model from overfitting to the training data.

Validation Split – The proportion of the dataset held back from training and used only for evaluating model performance during each epoch.

Max Workers – The number of parallel CPU worker threads used to load and preprocess images during training.

Image Size – The pixel resolution (width and height) images are resized to before being fed into the model.

Dropout – A regularization technique that randomly disables a fraction of neurons during each training step to reduce overfitting.

Stoch. Depth – The probability that any given ConvNeXt block is randomly skipped (dropped) during a training step, which forces the network to learn more robust features and acts as a strong regularizer. The torchvision defaults are 0.1 for ConvNeXt-Tiny and 0.4 for ConvNeXt-Small, reflecting that deeper models need more regularization. On a small dataset like headstamp images, lowering this value below the default (e.g. 0.05) reduces regularization pressure and lets the model fit the data more aggressively — useful if the model is underperforming. Raising it above the default increases regularization and can reduce overfitting, but push it too high and the model struggles to train at all because too many blocks are being skipped.

Use SWA – Enables Stochastic Weight Averaging, which averages model weights over multiple points late in training to produce a smoother, better-generalizing final model.

SWA Start – The fraction of total epochs at which SWA is forced to begin regardless of other conditions — a hard deadline that overrides patience-based waiting.

SWA Mode – Controls how SWA is triggered. In adaptive mode, the model waits for the minimum epoch, accuracy threshold, and patience conditions to be met before activating, rather than starting SWA at a fixed epoch.

Acc Threshold – The minimum training accuracy the model must reach before adaptive SWA can begin. Until this threshold is crossed, SWA will not activate regardless of other conditions.

Patience – In adaptive mode, once the accuracy threshold is met, the model watches for training accuracy to stall or decline. If that happens for this many consecutive epochs, SWA is triggered.

Min Epoch – The earliest epoch at which adaptive SWA is allowed to consider activating. The accuracy threshold and patience checks do not begin until this epoch is reached.

Use Focal Loss – Switches the loss function to Focal Loss, which down-weights easy, confidently-classified examples so the model focuses more of its learning on hard or rare cases.

Focal Gamma – The focusing parameter for Focal Loss. Higher values increase the penalty reduction on easy examples, making the model concentrate more heavily on difficult ones.

Use GPU (if available) – Instructs the trainer to use a CUDA-capable GPU for training if one is detected, which significantly accelerates computation compared to CPU-only training.

Full Dataset Training – Trains on the entire dataset without holding back a validation split. Typically used for a final production model run after hyperparameters have already been tuned.

OpenAI Api Models and Configuration

This version of the software includes a new feature that lets you leverage an LLM or OpenAI API compatible endpoint to classify your headstamps. This includes cloud-based models such as those offered directly by OpenAI (e.g. gpt-4.1), local models running in LM Studio (Qwen3-VL models work well), or the lightweight AI Case Sorter Server available at <https://github.com/sjseth/AI-Case-Sorter-Server>.

OpenAI API Configuration

Prompt

Classify this image. Respond with only the label in the following list which closely matches the Image. If the label is not in the list, return 'Unknown':

Here is the list:

{{headstamps}}

Managed Headstamps

Headstamp	
FC	Delete
Win	Delete
CBC	Delete
BLAZER	Delete
EXTREME	Delete
GFL	Delete
*	

Save Settings

Load Headstamps

Feed

API Endpoint URL

http://10.10.10.159:1234

API Key

Model (gpt-5, gpt-4.1, etc)

qwen3-vl-8b

Scale

100

Quality

80

Modified Size

31 KB

Win



Processing: 42ms
Compression: 1ms
Classification: 261ms

API Endpoint URL – The URL of the API endpoint to send classification requests to. Defaults to <https://api.openai.com> for OpenAI, but can be pointed at any OpenAI-compatible endpoint such as a local LM Studio instance or the AI Case Sorter Server.

API Key – The authentication key for the configured endpoint. Required for OpenAI and other cloud providers. Local endpoints such as LM Studio or the AI Case Sorter Server typically do not require one.

Model – The model identifier to use at the configured endpoint. For OpenAI this would be something like gpt-4.1; for LM Studio it is the model name as it appears in LM Studio. The model must be capable of processing images (vision-capable).

Prompt – The instruction sent to the LLM along with the headstamp image. The token {{headstamps}} in the prompt is automatically replaced with the current Managed Headstamps list before the request is sent, telling the model which labels it is allowed to return.

Managed Headstamps – The list of valid headstamp labels that will be injected into the prompt in place of {{headstamps}}. Labels can be added and removed manually, or populated automatically using the Load Headstamps button. These headstamps are required in run mode so you can configure the headstamp:slot relationships.

Scale – Resizes the image to the specified percentage of its original dimensions before sending to the API. A value of 100 sends the full 480x480 image and is recommended for best accuracy. Lower values reduce the payload size but can degrade the LLM's ability to read the headstamp.

Quality – JPEG compression quality applied to the image before sending to the API. A value of 100 is recommended; lower values reduce file size but can introduce compression artifacts that affect classification accuracy.

Modified Size – Displays the resulting file size of the image after the Scale and Quality settings have been applied, so you can gauge the tradeoff between image fidelity and network payload size.

Classification – Displays the label returned by the LLM after it processes the submitted image.

Processing – The time taken to capture and prepare the image for submission.

Compression – The time taken to apply the Scale and Quality compression to the image before sending.

Classification (timing) – The time taken for the API endpoint to respond with a classification result.

Save Settings – Saves the current API configuration including endpoint URL, API key, model, prompt, and headstamp list.

Load Headstamps – When using the AI Case Sorter Server, this button calls the server's /getheadstamps endpoint to automatically retrieve and populate the Managed Headstamps list with the classification labels from the currently loaded model, saving you from entering them manually.

Feed – Cycles a new case into the sorter, captures an image, and submits it to the configured LLM endpoint. The result is displayed in the Classification box. Use this to verify your configuration is working correctly before running a full sort session.

The Run Screen

Overview

The Run Screen is where you go to sort your brass. In this screen, you will select which slot where you want each headstamp to go. Unselected headstamps will go to the catch-all slot which is Slot #0. This slot cannot be configured.

Controls

The screenshot shows the Run Screen interface with various controls and settings. The interface is divided into several sections:

- Slot #0:** Contains a list of headstamps (AGUILA, CBC, GFL, PERFECTA, RP, WIN, XTREME) and their corresponding slot numbers (1, 2, 10, 18, 2). A checkbox for BLAZER is also present.
- Slot #1:** Contains a list of headstamps (Aluminum, BLAZER, XTREME, AGUILA, CBC, GFL, MAXXTech, RP, Nickle, PERFECTA, PMC, SPEER, WIN, WMA) and their corresponding slot numbers (24, 3, 2, 1, 21). A checkbox for BLAZER is also present.
- Slot #2:** Contains a list of headstamps (RP, PERFECTA, PMC, SPEER) and their corresponding slot numbers (21).
- Slot #3:** Contains a list of headstamps (RP, PERFECTA, PMC, SPEER) and their corresponding slot numbers (21).
- Top Controls:** Includes buttons for Start (5), Clear Slots (7), Reset Counters (8), Feed (9), Monitor (22), and a Store Images dropdown (21).
- Speed:** A slider control (11) for adjusting the speed.
- Confidence Floor:** A dropdown menu (12) for setting the confidence floor, currently set to 80%.
- Run Mode:** A dropdown menu (14) for selecting the run mode, currently set to Popular Highest Average.
- Sample Rotations:** A dropdown menu (15) for selecting the sample rotations, currently set to 4.
- Mode:** A dropdown menu (16) for selecting the mode, currently set to Serial.
- Package Mode:** A checkbox (10) for enabling package mode.
- Sorting Template:** A dropdown menu (26) for selecting the sorting template, currently set to Default.
- Total Run Count:** A display (13) showing the total run count, currently set to 34.
- Automatically Select Trays:** A checkbox (6) for enabling automatic tray selection.
- Use Parent Classifications:** A checkbox (23) for enabling parent classifications.
- Image Display:** A large image (19) showing a coin with the word "HORNADO" and "LUGER" visible. The classification is FC (17) and the confidence is 22% (18).

Here is a brief description of each of the controls on the run screen and their function.

- 1. Catch-All Slot (Slot #0)** – The first slot in the list is the catchall slot. Its function to serve as a place for all cases to go which are either not bound to another slot or are below the required **Confidence Floor** setting (#12)
- 2. Headstamp Counters** – In each slot configuration, for each headstamp there is an individual counter which is incremented whenever a case is dropped into that slot.
- 3. Slot Counter** – Each slot has a counter which is incremented when any case is dropped into that slot. It serves as a “Total” count for that slot.
- 4. Headstamp Checkbox** – This box must be checked to bind a headstamp to a slot. In the standard sorting mode, a headstamp can only be bound to one slot at a time. In package mode (#10), a headstamp can be

bound to multiple slots.

5. **Start Button** – This button starts the sorting process. Once the button is pressed, it becomes a “Stop” button which serves to stop the sorting process. If this button is grayed out, it is because the currently selected model has not yet been trained. Go back to the training screen and hit the “Train Model” button or select an already trained model from the Models menu.
6. **Automatically Select Trays** – When this is selected, the sorter will automatically bind an unbound headstamp to the first unbound tray. This will occur in a 1:1 fashion until all headstamps or all slots have been a headstamp binding. For example, if a CBC headstamp is detected but CBC has not been checked in any slot config, it will find the first slot which has no checkboxes checked and check the CBC box in that slot.
7. **Clear Slots** – This button clears the slot bindings.
8. **Reset Counters** – This button clears all the counters on the page including the headstamp, slot and total counters.
9. **Manual Feed** – This button is a manual “force feed” button and can be useful when you encounter an upside-down case which doesn’t trigger the proximity sensor. Also, at the end of the run it can be used to feed out the last 3 cases.
10. **Package Mode** – When checked the system switches from sorting mode to package mode. Package mode is designed to count a certain amount of brass into each slot. This is covered in more detail below.
11. **Speed** – The speed control allows you to slow down the system. This can be helpful if you are outrunning your brass feeder. It is also useful if you wish to have a bit of extra time to manually inspect the predicted results.
12. **Confidence Floor** – This value indicates the minimum confidence that a headstamp prediction must have to be sorted. Any brass with prediction lower than this confidence rating will be sent to the catch-all slot.
13. **Total Run Count** – This is the total number of cases which have been sorted since the last counter reset.
14. **Run Mode** – The sorter has 4 prediction modes which define the technique used to predict the headstamp. These are:

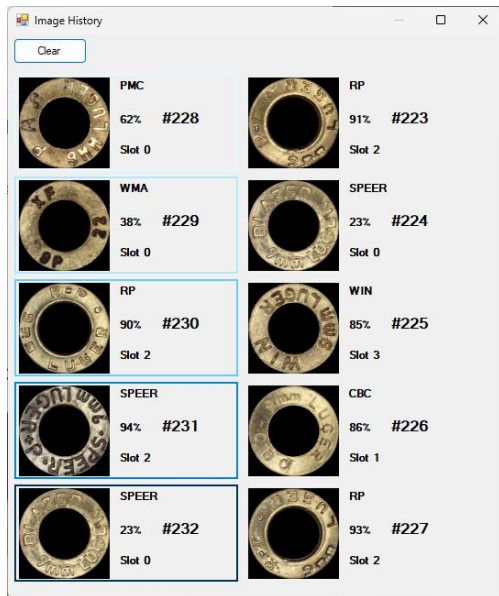
Single Image Highest Confidence (default) – This is the fastest run mode and uses a single prediction to classify the headstamp. With modern models, this is typically enough. For ConvNeXt models, this mode is now forced but the other modes will still be available for those upgrading from and using legacy models.

Multi Image Highest Confidence – When selected, the system will create a number of sample rotations (specified in #15) and predict each rotation. It will then select the prediction from the rotation with the highest confidence rating.

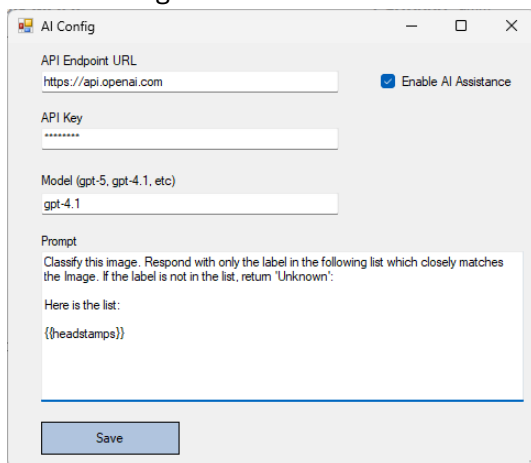
Highest Average Confidence – It will average the confidence ratings from the sample rotations and select the rotation with the highest average. For example, given 4 rotations and results like: FC: 70, WIN:75 FC:72, WIN:50 the system would select FC has the prediction with a prediction average of 71.

Popular Highest Average – It will average the confidence ratings from the sample rotations in addition to counting the number of predictions for each group. The largest group with the highest rating wins. For example, given 6 rotations with results like: FC: 50, FC:51 FC:55, WIN:60, WIN60, CBC:95, the system would select FC because it is the most popular in that it occurs 3 times. In this example if we removed one of the classifications for FC, then WIN would be the winner since the group size (population) for FC and WIN are the same and WIN has a higher confidence rating.

- 15. Sample Rotations** – The number of rotation images to create for each prediction. This option is not available in *Single Image Highest Confidence* run mode (#14)
- 16. Prediction Mode** – When predicting multiple rotations we have two modes. Serial mode predicts them one at a time (one after another) and Parallel mode predicts them all at the same time. Parallel mode will typically run faster than serial mode when running on a faster computer with many cores and memory. In Parallel mode, there will be a slight delay when starting the sort as it needs to create the number of processing threads based on the number of sample rotations selected (#15).
- 17. Headstamp Prediction** – This indicates which headstamp was predicted for a case.
- 18. Confidence** – Indicates the confidence for the headstamp prediction.
- 19. Processed Image Preview** – An image preview after all image processing has been applied.
- 20. Home Sorter Button** – If your sorter gets jammed during a run, you can stop the run and then hit this button to initiate a homing
- 21. Store Images** – This allows you to store images during the run. There are 4 modes for this option:
 - Above Confidence Floor – This can be used to capture your higher quality predictions
 - Below Confidence Floor – This can be used to capture those images you might want to use for more training
 - All Images – Captures everything
 - None (Default) – Capture no imagesWhen a capture mode is selected, you will see a link which takes you to the captured image store. In the application this is typically **C:\Program Files\SJSeth\AI Brass Sorter\training\images\{model id}_RUN**
- 22.** This opens a monitoring window which lets you see which images have been captured previously. The images will fill in from top to bottom, right to left and when out of space, continue back at the top left. Most current capture is outlined in black followed by graduating shades of blue for older captures.



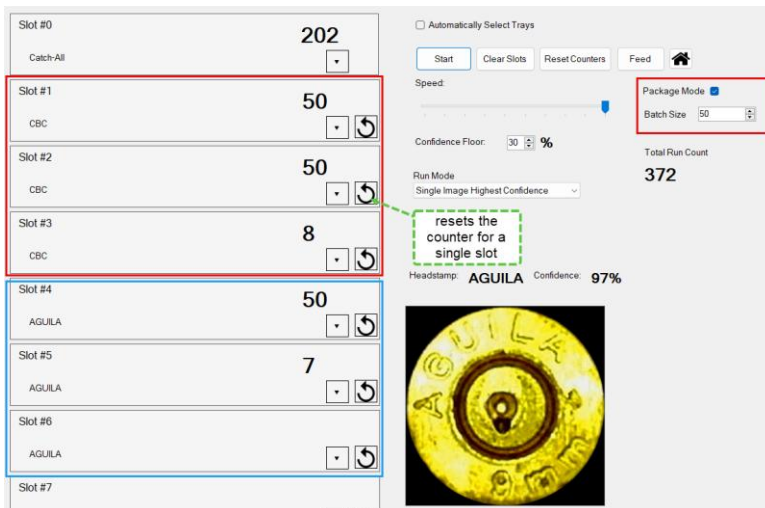
- 23. Use Parent Classifications** – If parent classifications have been defined for the active model, this checkbox will be displayed. Parent classifications are defined in the headstamp manager in the Models tab for a given model. When enabled, the slot configurations will show the parents and headstamps without parents. When parents are defined but not enabled, the headstamps will be grouped by the parents as seen above. (see pages 23 and 24 for more info on managing headstamps and parent classifications)
- 24. Parent Groups** – As covered previously in #23, if parents are defined for this model, these expandable/collapsible groupings will appear. Selecting a checkbox next to a group allows you to quickly select/deselect all the children in that group. (see pages 23 and 24 for more info on managing headstamps and parent classifications)
- 25. AI Fallback** – This feature allows you to use an AI Vision enabled model (such as chat gpt 5.x) to classify those images which fall below the confidence threshold. This link opens the configuration window. To disable/enable the feature check/uncheck the “Enable AI Assistance” box. See the section on OpenAI API Model configuration for more details



- 26. Sorting Templates** – This feature allows you to store template slot configurations within the same model. A template is basically a set of mappings between headstamps and slots and you can create templates to handle your various sorting needs.

Package Mode

In addition to sorting by slot, you can also use the system for sorting batches of brass for packaging. You simply need to enable package mode and specify the *batch size*. In package mode you can bind one or more headstamps to multiple slots. The system will fill each slot one at a time. When a slot has reached its capacity, the system will pause briefly and sound an alarm (3 beeps) to alert you before moving on to filling the next slot.

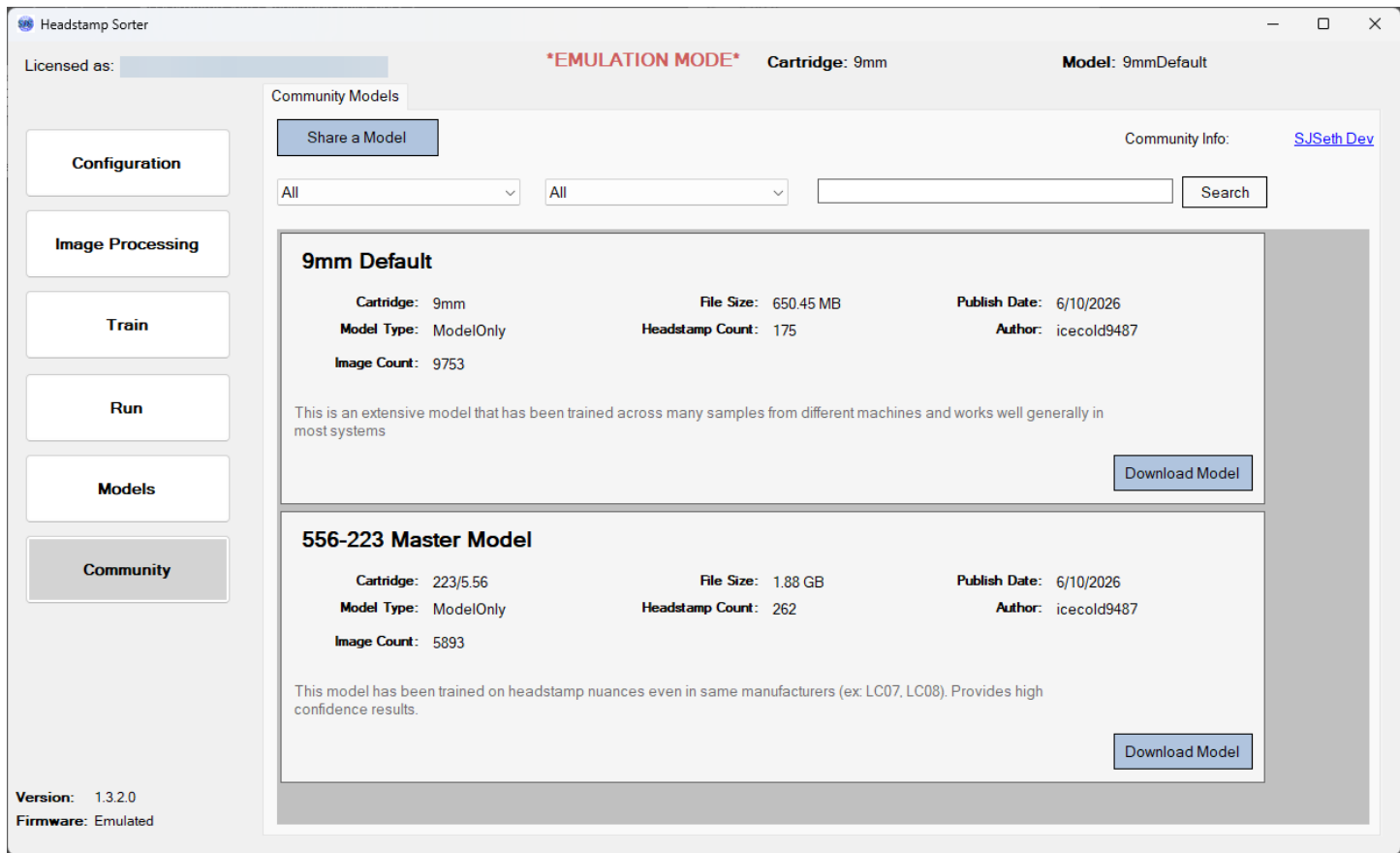


The screenshot above shows the system in package mode and configured to package two headstamps (CBC and AGUILA) in packages of 50. As you can see, two of the 3 bound slots for CBC are full and one of the Aguila slots is full. In this scenario, if monitoring the system, one could empty the bins for slot 1 and 2 into packages (boxes or bags) and then hit the reset button next to each slot. This will reset the counter for the slot making it available to package another batch. This can be done without stopping the system for continuous operation.

Community Features

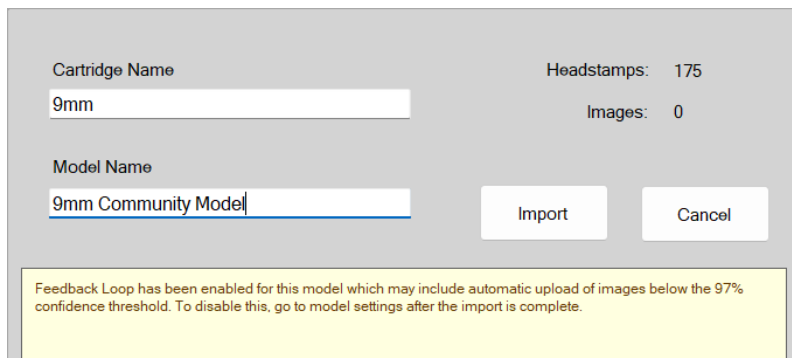
In version 1.3.2, there are new community features. This tab let's you access models which have been shared to the community. These models are managed an maintained by one or more model owners.

The models shared here are generally high quality and contain a large variety of samples. You can search for models here and quickly download/install them automatically by clicking the download button.



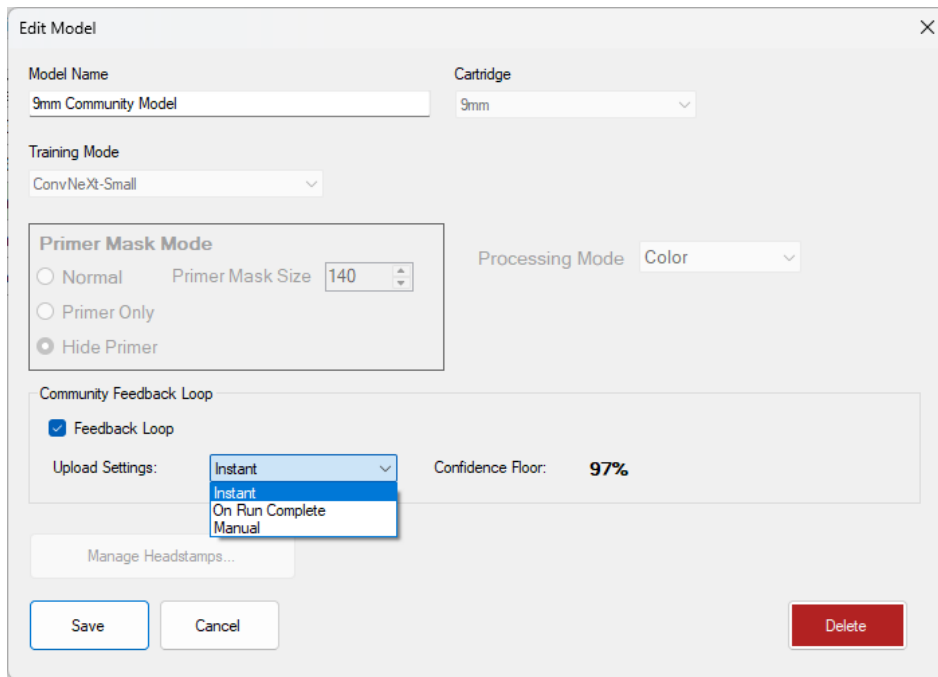
Feedback Loop

In the community models, often a feature named “Feedback Loop” will be enabled. This is a feature that the model publisher can enable so they can automatically receive images which are below their specified confidence threshold. When feedback loop is enabled, during a run cycle, if there is an available internet connection, any images falling below the threshold are uploaded to the server. The moderator can then choose to use those images to improve the community model. When downloading a model, you will be notified if this feature is enabled. See the yellow banner in the screenshot below.



If you do not want images to be automatically uploaded or want to turn off this feature, you can opt out of it in the model configuration after you have downloaded the model.

If you go to the model screen and edit the model, you will see some additional settings for the Feedback Loop feature

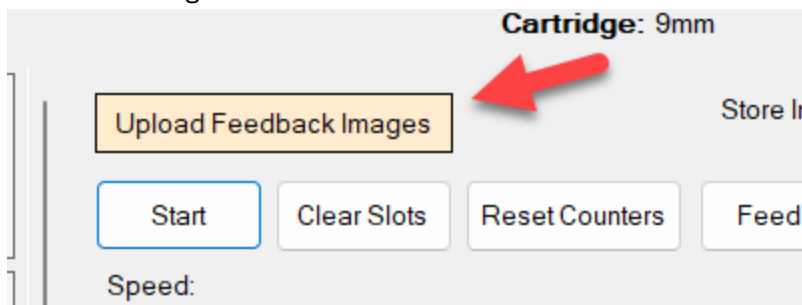


To turn off Feedback Loop for this model, simply uncheck the Feedback Loop checkbox. There are various modes available for the image upload.

Instant (Default) – During a run cycle if an image is below the defined confidence level, it is uploaded immediately via a background service which doesn't impede the run.

On Run Complete – Images are stored in a temporary folder and once the run completes they are uploaded to the server and then deleted from the folder.

Manual – Images are captured and stored in a temporary folder and the user is presented with an Upload Feedback Images button in the run screen.

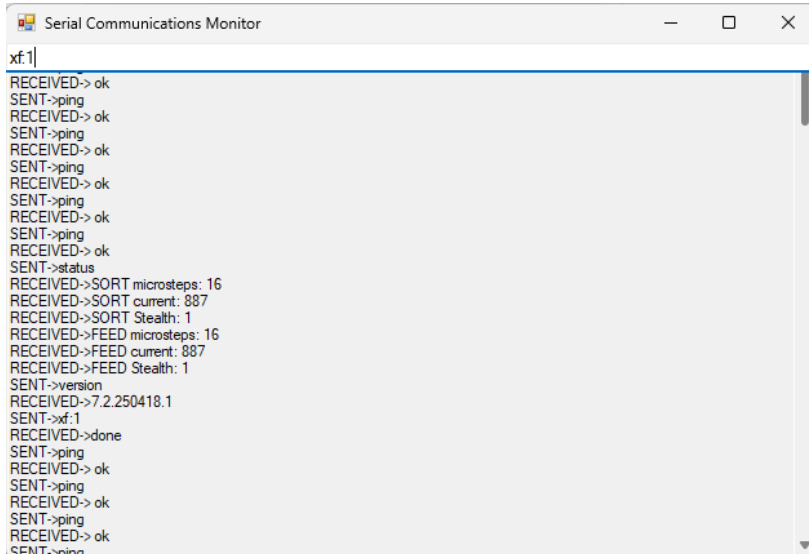


This button may also display in the other modes if there was a network issue which prevented an image from being uploaded during the run. To clear this indicator, you can simply navigate away from the run screen and then back again.

Advanced / Hidden Features

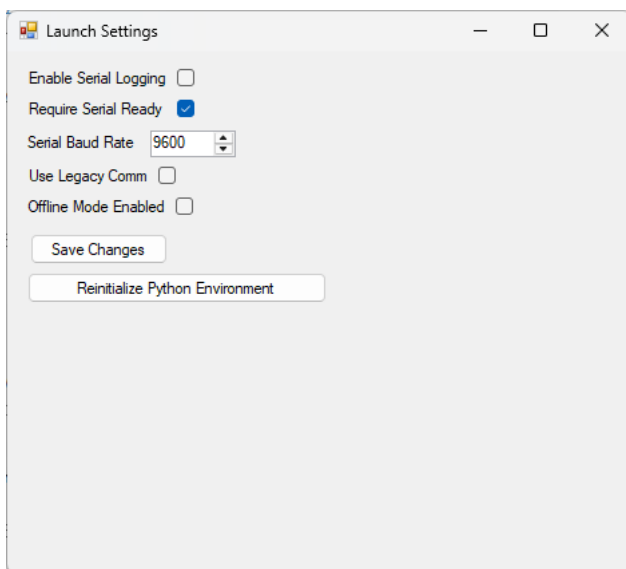
Serial Communication Monitor

You can launch a serial communication monitor using the keystroke combination **CTRL + K** which opens a window giving a report of incoming and outgoing serial messages between the app and the sorter. This can be seen in the following screenshot. Messages marked with SEND are from the software to the Arduino and those with RECEIVE are the responses from the Arduino to the software. In addition, as of 1.1.51, you can type and send commands manually in the text field above the monitor.



Launch Settings

The Launch Settings window can be accessed in the software with the key combination of **CTRL + S**. This contains the application initialization setting and should rarely need to be modified. Note that the software must be restarted after saving any changes before those changes take effect.



Enable Serial Logging – This feature logs serial communication to a file in the Data folder of the application. This can be useful for troubleshooting purposes or capturing communication for reverse engineering type applications.

Require Serial Ready – When connecting to the sorter, the software expects to receive a response of “Ready” from the sorter. If it does not receive a response within 3 seconds, the sorter will be ignored and most likely the software will go into Emulation mode. For custom implementations, you can disable this requirement.

Serial Baud Rate – The default value specified in the Arduino code is 9600. If modifications to the Arduino code have been made or a different sorter device is being used, this setting can be set in the software to restore communication.

Use Legacy Comm – This feature causes the software to continue communication over serial in the way it did prior to version 1.1.51. Only enable this feature if you have communication drops/timeouts or other issues after upgrading to 1.1.51 or greater.

Reinitialize Python Environment – If you want to change the mode for your python environment from GPU back to CPU or vice versa, use this option which will launch the configuration screen and let you select the appropriate mode.

Troubleshooting

Images are black/blank when running in emulation mode.

Check your Image Processing settings to make sure the image is within the crop area (if using manual or hybrid modes)

Sorter not detected; software goes to emulation mode.

There are several reasons for this, but the most common one is that the serial port is not available because it is in use by another application. Very typically that “other” application is the Arduino IDE. It is important to close the IDE as it can keep the serial port open and not allow the software to use it.

Another common issue is that the USB port is not connected or has a problem. You can try unplugging/replugging the USB cable or try connecting the USB cable to a different port. If you are using a USB hub, you can also try bypassing the hub and connecting the cable directly.

The Application Crashes when training starts after I press **Train Model**

This problem can occur for many different reasons, but these are the most common.

- If the CPU does not have AVX instruction support. (if you have already run a successful training previously, this is not the problem). The solution is to get a computer that supports AVX instructions.
- The processor is not x64 (64bit) architecture.
- The computer does not have an internet connection during the first training. When training a model for the first time, the software may attempt to download some of the external dependencies needed for the training. If there is no internet connection, the app may crash.
- The VS C++ redistributable was not installed. Solution is to install the redistributable from the software downloads page.
- Images have a headstamp mismatch. If you have manually renamed training images in windows explorer, you need to be sure that each classification (file prefix) also exists in the headstamp data for that model and carries the same case sensitivity. For instance, if you have a file like Fed111_23049820349823.jpg, you need to be sure your model also has a headstamp named Fed111. FED111 and fed111 are different headstamps.
- System out of memory. If your system has 8gb or less, or has other applications using resources, the system may not have enough memory to complete the operation. In this case, you should try rebooting your computer and on a fresh boot, launch the application and go directly to the training screen to run the train model operation.